Department of Electrical & Computer Engineering Technology (ECET)
School of Engineering, Technology, and Advanced Manufacturing (ETAM)

EET 4950

Senior Design Project

Safe Dip

A Safety-Oriented Hot-Dip Pick-and-Place System

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Abstract

Manual hot-dip coating operations expose workers to elevated temperatures, repetitive strain, and the risk of burns from hot, viscous materials. This project presents Safe Dip, a two-axis Cartesian gantry designed to automate the pick, dip, and place cycle while reducing direct operator exposure. The system provides controlled workpiece positioning, immersion depth, dwell time, and transfer between fixed input, dipping, and output stations.

An STM32 microcontroller coordinates the complete sequence through a cooperative, nonblocking control architecture. Horizontal motion is produced by a stepper-driven timing belt and linear rails, while a stepper-driven leadscrew controls the vertical axis. A conventional electromagnet handles ferromagnetic workpieces, a time-of-flight sensor detects their presence, and normally-closed limit switches provide homing and travel references. Serial and Modbus TCP communications report operating states, sensor information, and fault conditions.

Safety is treated as a primary design consideration. A hardwired Category 0 emergency-stop circuit removes power from the motor drivers independently of the microcontroller, while the electromagnet remains on a separate supply to avoid intentionally releasing a suspended workpiece during a motion stop. The laboratory prototype will be evaluated without a heated coating bath and will demonstrate the integration of the mechanical, electrical, sensing, control, communication, and safety subsystems. Supporting calculations evaluate the design's scalability toward the intended 10 kg payload.

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Chapter 1

Introduction

Summary

In this chapter, we introduce Safe Dip, a safety-oriented hot-dip pick-and-place system designed to remove operators from the hazardous environment of manual wax coating operations. We establish the motivation for automating this process, identify the risks associated with manual dipping, and review existing commercial solutions. We conclude by identifying the gap in the market that Safe Dip is designed to fill.

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1.1 Background and Problem Statement

Hot-melt dip coating, also referred to as hot dipping or hot-melt wax coating, involves submerging a part into a heated liquid medium such as molten wax, resin, or a similar coating compound. Upon removal, the liquid solidifies as it cools, forming a protective layer around the part that can be cleanly stripped away when needed. This process is commonly applied before long-term storage or bulk packaging to shield parts from corrosion, prevent contact damage that could dull sharp edges, and, in some cases, enhance a tool's grip or handling comfort [1], [2].

Safe Dip seeks to address this gap by providing an affordable, safety-conscious, and instrumented automation solution tailored to this class of hot-dip operations. It targets the small-to-medium production scale where off-the-shelf industrial robots are cost-prohibitive and general-purpose systems lack the domain-specific safety integration required by industrial electrical standards.

1.2 Need and Motivation

In facilities where this process is still performed manually, workers face sustained exposure to elevated temperatures, repetitive motion injuries, and the risk of burns or chemical contact. As soon as the weight of the objects surpasses a couple of kilograms, long-handled tongs or clippers are introduced, adding to the physical strain. Finally, most coating products are not only hot but also viscous, meaning that even the smallest splash can cause serious injuries. Together, these hazards make manual dipping one of the more accident-prone tasks on the shop floor.

Automating the dipping cycle would remove the operator from the hazard zone entirely, yet many smaller manufacturers have not adopted automated solutions due to cost barriers, system complexity, and the lack of scalable, application-specific equipment suited to their production scale.

In addition to improving operator safety, an effective automation solution should remain useful when the coating process changes. Many commercially available systems are designed around a particular chemical, workpiece geometry, or production workflow, making them difficult or expensive to repurpose. Safe Dip instead treats the dipping liquid as an external process variable: the system performs the same controlled pick, immersion, dwell, and placement sequence regardless of the specific coating, cleaning, or treatment medium being used. This chemical flexibility allows laboratories, prototype facilities, and small manufacturers to adapt to new formulations, suppliers, or applications without redesigning the automation equipment.

By combining this flexibility with a comparatively simple Cartesian architecture, Safe Dip seeks to address the gap between manual dipping and highly specialized industrial machinery. The objective is not to reproduce the throughput or complexity of a complete robotic coating cell, but to provide a lower-cost platform that improves safety, process repeatability, and adaptability at a scale accessible to smaller operations.

1.3 Similar Products and Comparison

Table 1.1
Comparison of commercial dip coating systems against Safe Dip

Product	**Price**	**Conveyor Feed**	**Variable Dip Depth**	**Pick & Place**	**Chemical Flexibility**
Safe Dip (Ours)	~$500-$600	YES	YES	YES	YES
Dip-Spin (EFFCO DS-250)	est. $15k–$25k	NO	NO	NO	YES
MS800 Tiltable Dip-Spin	est. $40k–$80k	NO	NO	NO	YES
Monorail Headstand Screw Machine	$2,999–$3,999	YES	NO	NO	Partial
Thread Locker Machine (TH-2004L3)	On Request	YES	NO	NO	Partial

Product	Price	Conveyor Feed	Variable Dip Depth	Pick & Place	Chemical Flexibility
Full Auto Thread Coater	$20k–$30k	YES	NO	NO	NO
Emmeti WAX-ON	est. $50k–$150k+	YES	YES	YES	Partial
South Shore Robotic Dip Cell	Custom Quote	NO	YES	YES	YES

Chemical flexibility is the ability of a system to operate with different coating, cleaning, and treatment liquids without requiring changes to its mechanical architecture or control logic. The system interacts with the process only through part detection, controlled handling, programmable immersion, and placement; therefore, the chemical composition of the liquid is treated as an external process variable rather than a machine-specific design input. Compatibility is limited primarily by the temperature tolerance and chemical resistance of the end effector and other exposed hardware, rather than by the operating principle of the system itself.

This separation between the machine and the dipping chemistry is important because it allows the same platform to be repurposed as process requirements, suppliers, or formulations change. Instead of designing dedicated equipment for each liquid or application, users can adapt the process by changing the dip medium and operating parameters while retaining the same automation hardware. This reduces dependence on a single chemical process, lowers the risk of future redesign, and makes Safe Dip more suitable for laboratories, prototyping environments, and small-scale manufacturing where process requirements may change frequently.

1.3.1 Aramfix EFFCO DS-250 Dip-Spin Machine

The EFFCO DS-250 is a batch-type industrial dip-spin coating machine designed for high-volume processing of small metal components such as fasteners, nuts, bolts, and stampings. The machine operates by loading parts into a perforated basket, submerging the entire basket into a coating tank, and then spinning the basket at high RPM to centrifugally remove excess

coating and achieve a uniform film thickness. This process enables extremely consistent results across thousands of parts per cycle, making it well-suited for mass production environments. The DS-250 features a heated coating tank and programmable dip and spin cycles, allowing operators to fine-tune coating thickness to specification. However, its batch-based design means individual part handling and positional control are not possible, as all parts in the basket are processed simultaneously without differentiation. The machine requires manual loading and unloading of the basket, meaning human intervention is necessary at each production cycle. Its industrial footprint and price range of $15,000 to $25,000 place it well beyond the reach of small-scale or research-oriented operations. Unlike Safe Dip, the DS-250 cannot adapt to varying part geometries within a single run or interact with a conveyor-based workflow.

1.3.2 MS800 Tiltable Dip-Spin Machine

The MS800 is a fully automatic, tiltable dip-spin coating machine designed for heavy-duty industrial production of fasteners and small precision components. Its distinguishing feature is a motorized tilting mechanism that repositions the basket during the drain cycle, improving coating drainage and reducing waste compared to standard vertical-axis spin machines. The system is PLC-controlled, offering precise regulation of dip duration, spin speed, tilt angle, and temperature, resulting in coating uniformity that is difficult to match with manual or semi-automatic methods. With a throughput capacity in the thousands of parts per hour, the MS800 is optimized for large-scale manufacturing lines rather than flexible or small-batch applications. The machine's price range of $40,000 to $80,000 reflects its industrial-grade construction and automation level. Like other dip-spin systems, it is fundamentally a batch processor and cannot handle individual components, vary dip depth per part, or interface with a pick-and-place conveyor system. Its complexity also demands trained operators and dedicated maintenance protocols. While it represents the high end of coating machine capability, it addresses a fundamentally different operational context than Safe Dip.

1.3.3 Monorail Headstand Screw Coating Machine

The monorail headstand screw coating machine is a continuous-flow system designed specifically for the automated surface treatment of headed fasteners such as screws, bolts, and rivets. Parts are loaded onto an overhead rail conveyor by their heads, hanging inverted as they travel through a dip station where their threaded sections are coated in the desired chemical or lubricant. The continuous rail feed allows throughput rates of 10,000 to 20,000 parts per hour, making it one of the most efficient solutions for large-volume screw coating applications. Because parts self-hang from their heads, no gripper or pick-and-place mechanism is

required, which simplifies the mechanical design significantly. However, this same feature makes it incompatible with any component that lacks a head or flange to hang from, limiting its applicability to a narrow category of fastener geometry. The coating type per production run is fixed, and changing the chemical requires a full tank flush and reconfiguration. Priced between $2,999 and $3,999, it is the most affordable automated option reviewed, but its single-purpose nature and geometric constraints make it unsuitable for the diverse component handling that Safe Dip is designed to support.

1.3.4 Thread Locker Coating Machine — TH-2004L3

The TH-2004L3 is a purpose-built automatic dispensing machine designed to apply thread-locking adhesive, such as Loctite or Vibra-Seal equivalents, directly onto the threads of screws and bolts prior to assembly. The machine integrates a bowl feeder for bulk part loading, a precision dispensing head that applies a metered quantity of adhesive to each thread zone, and a conveyor exit for processed parts. Its dispensing mechanism is calibrated to apply only to the thread region, avoiding contamination of the head or shank, which is critical for assembly line compatibility. The system handles fastener sizes from M3 to M36 and operates at high throughput, making it appropriate for automotive, electronics, and aerospace assembly environments. Priced between $3,300 and $4,500, it sits at the lower end of the industrial machine spectrum, though it is still an order of magnitude more expensive than Safe Dip. Its fundamental limitation is that it is chemically and geometrically constrained: it applies a single fluid type to a single class of headed fasteners, with no capability to handle other part geometries or coating materials. It also has no pick-and-place functionality and cannot interface with a generalized conveyor workflow.

1.3.5 Full Automatic Thread Coater

The full automatic thread coater is a high-capacity, end-to-end automated machine designed for the uninterrupted production-line application of thread-locking compounds or sealants on screws and bolts. It integrates automated part feeding, precision coating application, and a discharge conveyor into a single self-contained unit, eliminating the need for manual intervention during normal operation. The system is optimized for single-fluid, single-geometry workflows and is typically configured for a specific fastener size range at installation. Throughput is very high, and the machine is engineered for continuous multi-shift operation in demanding industrial environments. Priced between $20,000 and $30,000, it represents a significant capital investment best justified in high-volume production contexts. Its automation level, while impressive within its domain, comes at the cost of flexibility: changing the coating fluid or accommodating a different part geometry requires machine reconfiguration

and downtime. It lacks any conveyor-integrated pick-and-place capability and cannot adjust dip depth or dwell time per individual part. These constraints make it an effective solution for its intended use case but a poor match for the adaptive, multi-geometry coating scenario that Safe Dip is designed to address.

1.3.6 Emmeti WAX-ON Machine

The Emmeti WAX-ON is a fully robotic, end-to-end bottle waxing line developed for high-volume industrial winery and spirits production. The system integrates a depalletizer, an infeed conveyor, a robotic dipping arm, a multi-tank wax station with interchangeable color tanks on rails, and a vision-based defect inspection camera, all feeding into an automated case packer. Its robotic arm performs true pick-and-place operations, selecting individual bottles from the conveyor and submerging them into the wax tank with consistent depth and dwell time control. The 3D part identification system allows the machine to adapt to minor variations in bottle positioning without human correction. Coating uniformity is achieved through a combination of precise dip speed, controlled withdrawal rate, and optional rotation of the bottle during extraction. The system is capable of processing hundreds of bottles per hour with minimal operator involvement, making it one of the most capable waxing machines commercially available. However, its price range of $50,000 to $150,000 and its large industrial footprint place it entirely outside the scope of small-scale or research applications. It is also purpose-designed for bottles and is not adaptable to other component geometries without significant engineering modification.

1.3.7 South Shore Controls Robotic Dip Cell

The South Shore Controls robotic dip cell is a custom-engineered industrial automation solution designed for precision dipping, coating, and chemical treatment of components across a range of industries, including food processing, medical device manufacturing, and industrial finishing. The system is built around a 6-degree-of-freedom industrial robot arm that provides full spatial control over the component throughout the dip cycle, enabling complex dip trajectories, variable dip angles, and multi-stage coating sequences that rigid gantry systems cannot replicate. Custom end-effectors, or grippers, are designed to match the specific geometry of the target component, ensuring reliable handling even for irregular or delicate parts. The system is PLC-controlled and fully programmable, with PID-regulated temperature control on the coating tank and real-time process monitoring. Pricing is entirely quote-based and typically falls in the range of $50,000 to $100,000 depending on configuration, making it accessible only to well-capitalized industrial clients. Unlike Safe Dip, it does not incorporate a conveyor-based intake and exit workflow; parts must be presented to the robot

by separate handling systems. Its strength lies in precision and flexibility within a controlled cell environment, rather than inline conveyor integration.

1.3.8 Other Alternatives

Before arriving at the proposed design, several alternative approaches were evaluated across three key areas of the system: the overall mechanical architecture, the end effector mechanism, and the fail-safe strategy for power loss events. Each alternative was assessed against the project's core priorities of operator safety, cost, mechanical simplicity, and suitability for the hot-dip environment. The following subsections document the options considered and the reasoning behind the decisions made.

1.3.8.1 Mechanical Architecture

From the very beginning of the project, the idea of a gantry competed against the idea of a robotic arm, as the one in Fig. 1.1. While it presented enough dexterity for the task at hand, a couple of issues remained, regardless of the degrees of freedom it presented:

- The bigger the number of degrees of freedom it had, the bigger the electronic and computer complexity the system would carry, as each degree of freedom would require at least an additional actuator.
- Mechanically, the fact that the arm is only anchored at one point at the bottom, offset from the object's position, meant that the motors would have to generate and support the entirety of the weight of the object.
- Given the height of some hot-dipping pots, the arm would either require unusually long sections, which would toll even more the motors to sustain the torque, all this to be able to get deep enough for the pot, and high enough to avoid it while grabbing the object.

In contrast, the mechanics of a cartesian gantry are pretty straightforward: only a motor per axis is required, and the positioning of them makes it such that the structure bears most of the weight in a predictable way.

Finally, a last alternative was considered where the objects being dipped could just move through a conveyor belt and have the liquid poured from the top, but that was not acceptable, since materials like hot wax would harden and break the belt, and a uniform coating would be hard to achieve.

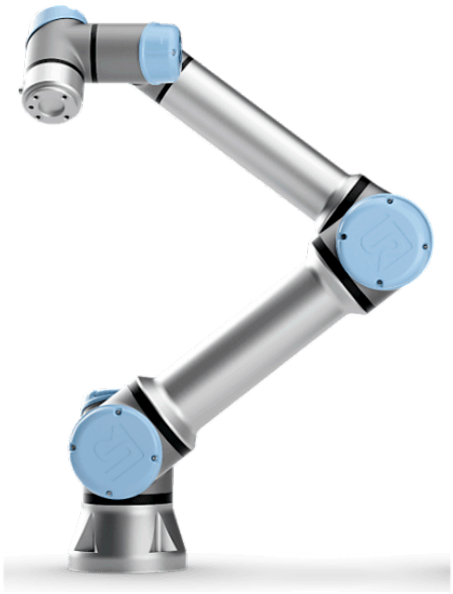


Fig. 1.1: Universal Robots UR5e Collaborative Robot Arm

1.3.8.2 Effector Alternatives

Another aspect worth consideration is the end effector (i.e., the mechanism that grabs the objects). While a claw is usually the first option everyone thinks of, it also comes with its own disadvantages:

- Claws must have a strong enough grip to ensure the object doesn't slip, while being soft enough as to not deform it or break it.
- The grip strength is dependent on the geometry of the object and the positioning of the claw.
- As the weight of the objects to be picked increases, it's been recommended to use a pneumatic claw, which introduces a new design dimension, complications, and considerations.

While an electro-permanent magnet would have been a more suitable effector, more aligned with the safety-conscious objective of this project, its availability is limited, and its price is cost-prohibitive for a prototype.

1.3.8.3 E-Stop Alternatives

Finally, for a project created out of the need for a safe alternative to manual operation, a lot of thought has been put into how the system should behave in the case an emergency stop was required. International standard IEC 60204-1 specifies three categories of emergency stops [3]:

- Category 0: uncontrolled stop by immediate removal of actuator power.
- Category 1: controlled stop, with power retained long enough to stop, followed by power removal.
- Category 2: controlled stop with actuator power remaining available.

Category 1 requires more complexity than category 0, since it requires hardware to disconnect power after a specific time. While stepper motors could benefit from a soft deceleration, given the speed they reach, the complexity outweighed the benefits for this project. Furthermore, the microcontroller cannot be used to trigger a power-off sequence; it is always assumed that it can fail too.

Category 2 simply didn't fit the requirements of this project.

1.4 System Requirements and Success Criteria

Each requirement below describes a specific, observable behavior with a defined pass condition. The system is treated as a black box for these requirements, and emphasis is placed on interactions with system boundaries, the required level of performance, and physical requirements.

Table 1.2
System requirements and verification criteria

ID	Requirement	Test
R-01	The system shall not initiate motion unless a workpiece is detected at the input station.	Place a workpiece at the input station and command an operating cycle. The test passes if the system initiates the pick sequence. Remove the workpiece and command another cycle. The test passes if the system does not initiate motion.
R-02	The system shall position the end effector at the programmed pickup position within ± 1 cm along both axes before acquiring the workpiece.	Command the system to the programmed pickup position and measure the position of the end effector relative to the target. The test passes if the measured error is within ± 1 cm along both axes.
R-03	The system shall retain the workpiece throughout the complete pick-dip-place cycle without slipping or dropping it.	Execute one complete pick-dip-place cycle using the approved prototype payload. The test passes if the workpiece does not visibly slip or detach at any point during the cycle.
R-04	The system shall lower the workpiece to the programmed immersion depth within ± 1 cm.	Execute one dip sequence and measure the actual immersion depth relative to the programmed value. The test passes if the measured depth is within ± 1 cm of the programmed depth.

ID	Requirement	Test
R-05	The system shall hold the workpiece at the programmed immersion depth for the programmed dwell time within ± 1 s.	Execute one dip sequence and measure the dwell period with a stopwatch or equivalent timing instrument. The test passes if the measured dwell time is within ± 1 s of the programmed value.
R-06	The system shall complete one full unattended pick-dip-place cycle, beginning with workpiece detection at the input station and ending with workpiece release at the output station and return to the home position.	Place a workpiece at the input station and command one operating cycle. Confirm that the system detects and acquires the workpiece, transfers it to the dipping position, lowers it to the programmed immersion depth, holds it for the programmed dwell time, raises it, transfers it to the output station, releases it, and returns to the home position. The test passes if every stage is completed in the correct order without operator intervention.
R-07	The system shall remove power from both stepper-motor drivers when the emergency stop is actuated.	Actuate the emergency stop while an axis is moving and measure the power supplied to both stepper-motor drivers. The test passes if driver power is removed and all axis motion stops.

ID	Requirement	Test
R-08	The system shall remain inhibited following an emergency-stop event until the emergency-stop button has been reset and the operator performs a deliberate restart action.	Actuate the emergency stop during operation, reset the emergency-stop button, and attempt to command motion without performing the restart action. The test passes if no motion occurs. Perform the deliberate restart action and confirm that normal operation can be re-established.
R-09	The system shall transmit its current operating state and any active faults through the selected communications interface during operation.	Monitor the communications interface during one complete operating cycle and introduce at least one detectable fault condition. The test passes if each operating-state transition and the active fault are transmitted with the correct labels.
R-10	The system shall complete ten consecutive unattended full pick-dip-place cycles using the approved prototype payload without dropping a workpiece, missing the intended placement position, or requiring operator intervention between cycles.	Command the system to perform ten consecutive full pick-dip-place cycles using the approved prototype payload. Record the cycle time, commanded immersion depth, measured immersion depth, and measured dwell time for each cycle. The test passes if all ten cycles are completed, no workpiece is dropped, no placement is missed, and the measured immersion depth and dwell time remain within their specified tolerances.

ID	Requirement	Test
R-11	The system shall complete the full pick-dip-place cycle while carrying a workpiece with a mass of 1 lb.	Attach a 1 lb workpiece and command one complete pick-dip-place cycle. The test passes if the system lifts, transports, lowers, retains, and releases the workpiece without stalling, losing position, slipping, or dropping it.

1.5 Operating Limitations

The following constraints define the boundaries within which Safe Dip is designed to operate, which are outside the control of the system. They are not subject to pass/fail testing, but scope the design and inform component selection.

- Safe Dip operates along two axes only: a horizontal axis for transfer between the input station, dipping bath, and output station, and a vertical axis for picking, dipping, and placing the workpiece. The system does not support rotation or movement perpendicular to the horizontal travel path.
- The system is designed to handle cylindrical workpieces within the specified size and payload limits.
- Workpieces must be presented at a known, fixed input position. The system does not perform lateral alignment, orientation correction, or vision-based localization.
- Each workpiece must provide a clean, flat, ferromagnetic attachment surface that remains outside the required coating area. Because the electromagnet remains in contact with this surface throughout the cycle, the attachment area will not be dipped or coated. Workpieces requiring complete surface coverage would require a different end effector or a secondary coating operation.

For the laboratory prototype, the following additional operating limitations apply:

- The laboratory prototype is designed and validated for workpieces with a maximum mass of 1 lb. Operation above this payload is outside the prototype's validated operating range.
- The prototype shall not be operated with a heated dipping bath or exposed to temperatures beyond the rated limits of its selected components. All acceptance testing will be performed using an unheated, laboratory-safe medium under room-temperature conditions.

- The prototype does not validate operation near molten wax, heated resin, process fumes, splashing hot material, or other industrial hot-dip conditions.

This limitation keeps the prototype safe and practical to demonstrate in the laboratory. Using molten wax or another heated material would introduce burn, splash, spill, and fume hazards that would require additional protective equipment and facilities. Heat could also affect the frame, wiring, sensors, motors, lubrication, and end effector.

Although these issues would need to be addressed in an industrial version, detailed thermal and mechanical design is outside the main scope of the ECET prototype. The project therefore focuses on demonstrating the electrical, control, sensing, communication, and safety systems at room temperature.

1.6 System Block Diagram

The Safe Dip system can be divided into 7 modules as shown in Fig. 1.2.

The power distribution module is composed of the power supply and is in charge of converting wall power into the different types of voltage needed.

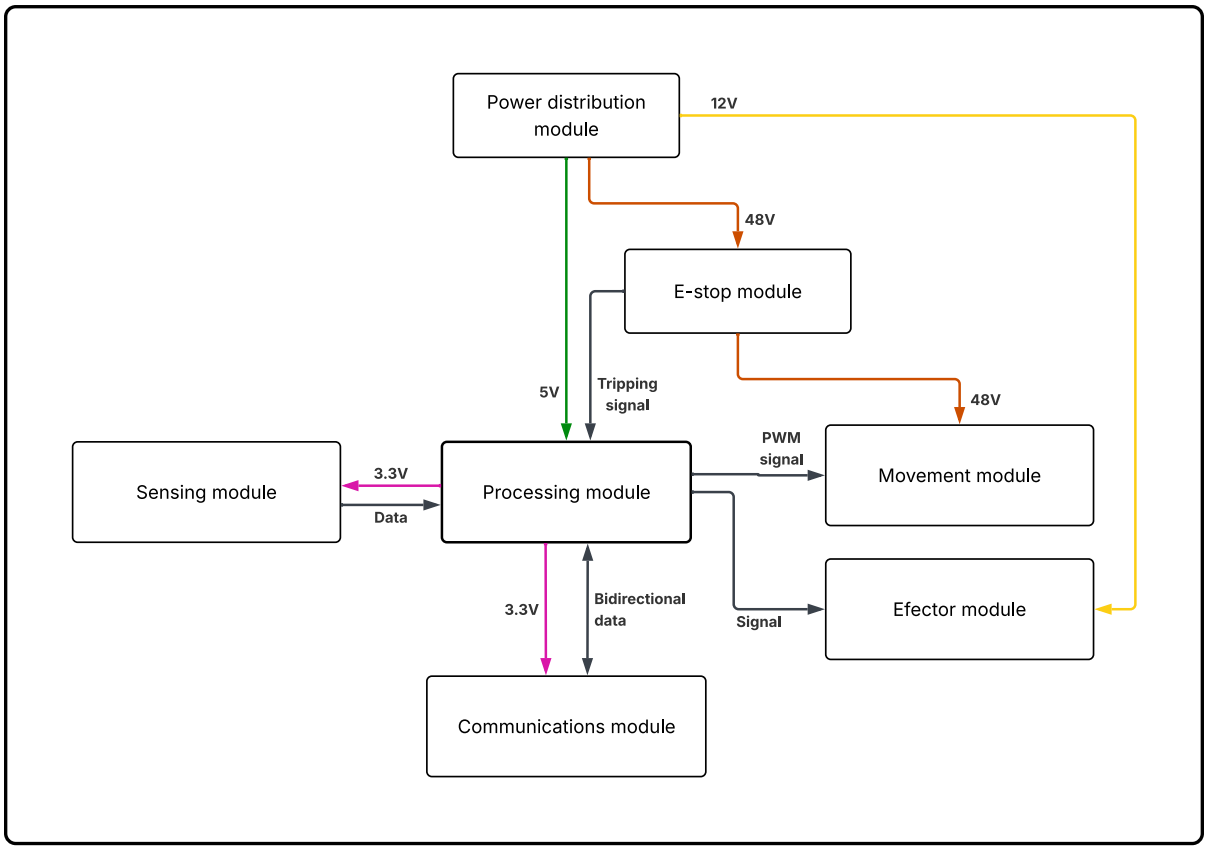


Fig. 1.2: System-wide block diagram

The E-stop module will contain all the mechanisms needed to perform a category-0 emergency stop, able to cut the power to the movement module. It is also in charge of alerting the processing module about the E-stop so that it can handle it gracefully.

The processing module is the heart of the system, encapsulating the microcontroller, which receives all inputs and translates them into the corresponding actuator responses. It also includes any and all components to ensure that the incoming data is at an acceptable input voltage for the microcontroller. This module receives its power from the power distribution module, and through the microcontroller's on-board power converter, it feeds 3.3 V to the other modules.

The sensing module is composed of all the time-of-flight (TOF) sensors used in this project, as well as limiting switches and other sensors.

The movement module is composed of all the stepper motors and controllers required for the horizontal movement of the gantry and the movement of the vertical assembly.

The effector module controls the part of the system that grabs the object and is composed of an electromagnet. It is nonetheless designed to be as modular as possible, so that a change of effector type is not hard to implement.

The communications module is an addition to the processing module in case communications are decided to be done via Ethernet TCP-Modbus. It would include a module capable of Ethernet communications.

The functional modules described above are expanded in Fig. 1.3 to show the electrical implementation of Safe Dip and the interfaces between its major components. The diagram identifies the system's power-distribution paths, control signals, sensor connections, communication interfaces, and actuator drivers. It also distinguishes between the high-power circuitry used by the motion and end-effector subsystems and the low-voltage circuitry used by the processing, sensing, and communications modules. This hardware-level representation provides the basis for selecting components, defining subsystem specifications, preparing the power budget, and verifying that the modules can operate together as an integrated system.

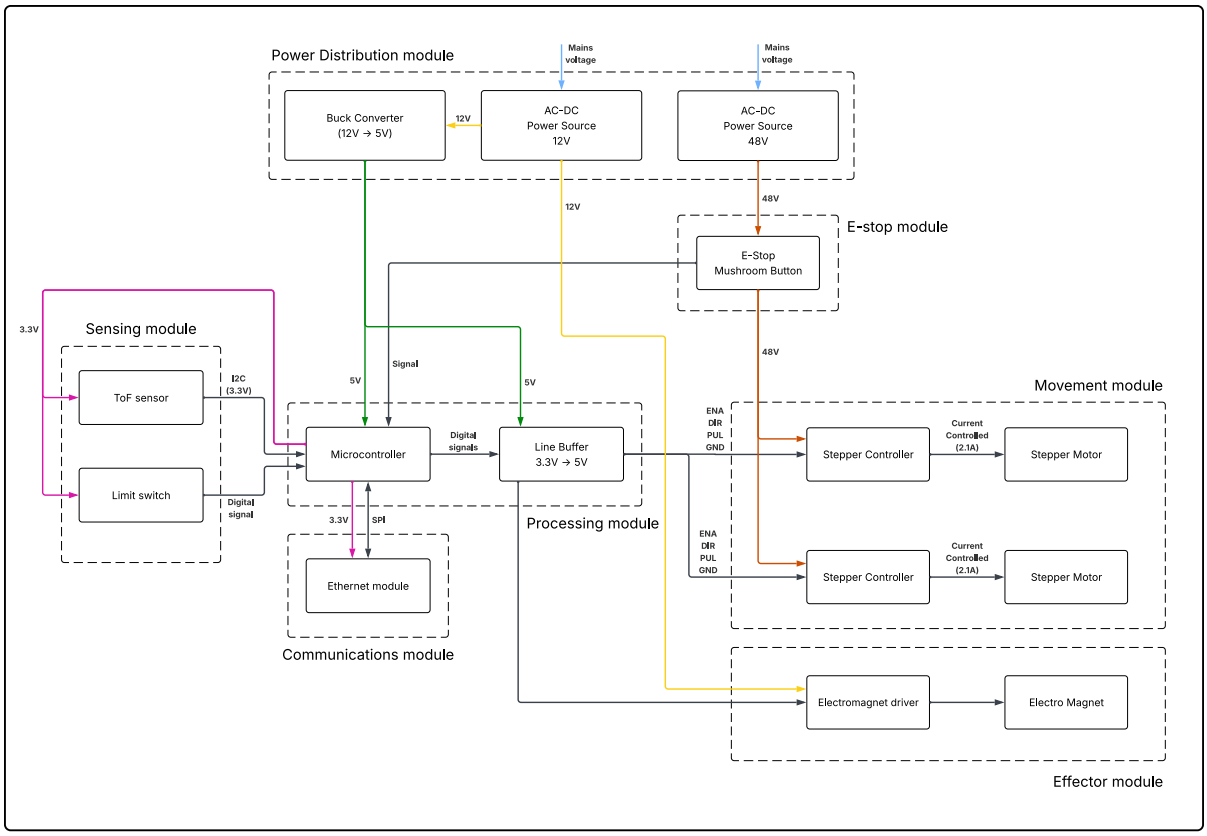


Fig. 1.3: Hardware architecture

1.7 Engineering Specifications

1.7.1 Power Distribution

Table 1.3
Power distribution specifications

Specific Component	Engineering Specifications	Justification & Verification
AC-DC power supply (motors)	• Input: 110 VAC, 60 Hz • Output: 24 VDC • Must supply sufficient current for two stepper motors at peak load	Verified by measuring output voltage under full load with a multimeter.
AC-DC power supply (12 V)	• Input: 110 VAC, 60 Hz • Output: 12 VDC • Ripple and noise within acceptable limits for logic, sensors, and effector	Verified by measuring output voltage and ripple with an oscilloscope under full load.
DC-DC Buck Converter (12 V $\longrightarrow$ 5 V)	• Input: 12 VDC • Output: 5 VDC • Ripple and noise within acceptable limits for logic and sensors	Verified by measuring output voltage and ripple with an oscilloscope under full load.

1.7.2 E-Stop

Table 1.4
E-stop specifications

Specific Component	Engineering Specifications	Justification & Verification
Emergency stop button	• Must offer a Normally Closed channel for cutting the power to the motors • Must be able to handle peak power coming from the AC-DC power supply (motors) • Requires manual reset before power is restored	Verified by actuating the E-stop during operation and confirming output power is cut.

1.7.3 Processing

Table 1.5
Processing specifications

Specific Component	Engineering Specifications	Justification & Verification
Microcontroller	• Operates at 3.3 VDC • Sufficient GPIO for stepper control, sensor inputs, and effector output • Supports USB UART for baseline serial communications • Capable of driving a Pulse Train Output (PTO) at the desired frequency set by the stepper and stepper controller	Verified by using an oscilloscope to check GPIO voltage, and an oscilloscope to verify PTO frequency.

1.7.4 Sensing

Table 1.6
Sensing specifications

Specific Component	Engineering Specifications	Justification & Verification
Time of Flight (TOF) sensor	• Operating voltage: 3.3 VDC • Detection range sufficient to cover the input station • Blind zone under 3 cm • Communication protocol compatible with the microcontroller	Verified by placing and removing a part at the input station and confirming correct detection in both cases.
Limit switches	• Normally-closed configuration • One per axis end, used for homing • Rated for the operating environment	Verified by commanding each axis to its respective home and confirming motion stops upon switch actuation.

1.7.5 Movement

Table 1.7
Movement specifications

Specific Component	Engineering Specifications	Justification & Verification
Stepper motors (x2)	• NEMA 23 frame • Holding torque sufficient to lift rated payload on vertical axis and drive belt under load on horizontal axis • Positional accuracy: ±1 cm	Verified by commanding each axis to a known target and measuring offset with a ruler.
Stepper motor drivers (x2)	• Compatible with a 24 VDC supply • Microstepping support • Current output set to motor rating	Verified by confirming smooth motion on both axes across the full travel range.

1.7.6 End Effector

Table 1.8
End effector specifications

Specific Component	Engineering Specifications	Justification & Verification
Electromagnet	• Holding force sufficient to retain rated payload throughout the dip cycle • Controlled by a single digital output from the microcontroller	Verified by attaching a part at rated payload and confirming no drop occurs throughout one full pick-dip-place cycle.
MOSFET driver	• Logic-level gate compatible with the microcontroller's 3.3 V or 5 V output • Drain-source voltage rating greater than the 12 V electromagnet supply with adequate transient margin • Continuous drain-current rating greater than the maximum electromagnet current • Low on-state resistance at a gate voltage	Command the electromagnet on and off while monitoring the gate, drain, and electromagnet voltage with an oscilloscope. The test passes if the MOSFET switches the full electromagnet current reliably, remains within an acceptable temperature range, and no damaging voltage transient is observed when the electromagnet is de-energized.

1.7.7 Communications

Table 1.9
Communications specifications

Specific Component	**Engineering Specifications**	**Justification & Verification**
Ethernet module	• Operating voltage compatible with the processing module • SPI interface compatible with microcontroller • Reports system state, axis positions, and fault events • Supports Ethernet and TCP socket communication	Verify communication between the STM32 and Ethernet module through SPI. Establish a TCP connection from an external computer and confirm that data can be transmitted and received.

Chapter 2

Background Research

Summary

This chapter describes the theory and the design of Safe Dip in detail. We explain the system's two-axis Cartesian gantry architecture and define its operating limitations. We present the testable system requirements and their verification criteria, describe each functional module through a system block diagram, provide engineering specifications for each subsystem, and outline the project budget.

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2.3 Software and Programming Languages 36
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2.1 Concepts, Algorithms, and Equations

This section presents the theory behind the design: how the axes convert motor rotation into motion, how the firmware generates and shapes the step pulses, and how the electrical loads were sized.

2.1.1 Axis Motion and Resolution

With the stepper drivers set to P microsteps per revolution, the leadscrew nut moves

$$\Delta x = \frac{L}{P} \tag{2.1}$$

per step, where L is the leadscrew lead. At a linear speed v the step rate is

$$f_{\text{step}} = \frac{v}{\Delta x} \tag{2.2}$$

$$m_{\text{safe}} = \frac{2\pi T(\omega)\eta}{SLg} \tag{2.3}$$

Here $T(\omega)$ is the motor torque at the operating speed, η is the mechanical efficiency of the leadscrew, S is the safety factor, and g is the gravitational acceleration. The derivation of this expression from the leadscrew kinematics is given in Appendix D.

2.1.2 Step Pulse Generation

Each axis uses a hardware timer channel in output-compare toggle mode, so the pin changes level every half period. With a timer clock f_{timer}, the compare register is advanced by

$$N_{\text{half}} = \frac{f_{\text{timer}}}{2 f_{\text{step}}} \tag{2.4}$$

timer ticks on every event.

2.1.3 Motion Profile

The motion planner accelerates at a constant rate a, cruises at the maximum speed, and brakes down to a low landing speed v_{land} so that the axis stops exactly on the target step. Braking starts when the remaining distance equals

$$d_{\text{brake}} = \frac{v^2 - v_{\text{land}}^2}{2a} \tag{2.5}$$

For a move of D steps too short to reach the maximum speed, the peak speed is

$$v_{\text{peak}} = \sqrt{aD} \tag{2.6}$$

2.1.4 Electrical Loads

The power drawn by each load is $P = VI$, and the total is compared with the rated power of each supply. The magnet current follows from its measured resistance R,

$$I = \frac{V}{R} \tag{2.7}$$

2.2 Hardware Components and Mechanical Design (CAD)

The components selected for Safe Dip were chosen to satisfy the engineering specifications defined in Section 1.7 while remaining within the project's budget constraint and accessible through standard commercial channels. The hardware is divided into two categories: the electronics responsible for control, motion, sensing, and communications, and the structural components that form the physical frame of the gantry.

2.2.1 Mechanical structure

The structural frame and motion transmission of Safe Dip are built from standard commercial extrusion and linear-motion components, selected so the dimensions listed in Table 2.1 can be met without custom machining.

The frame is assembled from 4040 T-slot aluminum extrusion, which forms the horizontal beam, the vertical column, and the supporting legs shown in Fig. 2.1. The T-slot profile allows components to be mounted at arbitrary positions along each beam and secured with T-nuts and M5 hardware, which simplifies alignment during assembly and permits adjustment of the input, dipping, and output station positions.

The horizontal axis is guided by an MGN12H linear rail and carriage and driven by a GT2 timing belt, converting the rotation of the horizontal NEMA 23 into the linear travel between stations described in Table 2.1. The vertical axis uses a Tr8x2 leadscrew supported by pillow-block bearings and coupled to the vertical NEMA 23 through a flexible shaft coupler; the leadscrew converts motor rotation into the controlled lifting and immersion motion, and its selection is supported by the leadscrew speed and payload calculations in Appendix A. An anti-backlash nut block is used on the leadscrew to remove axial play so that programmed immersion depth remains repeatable across cycles.

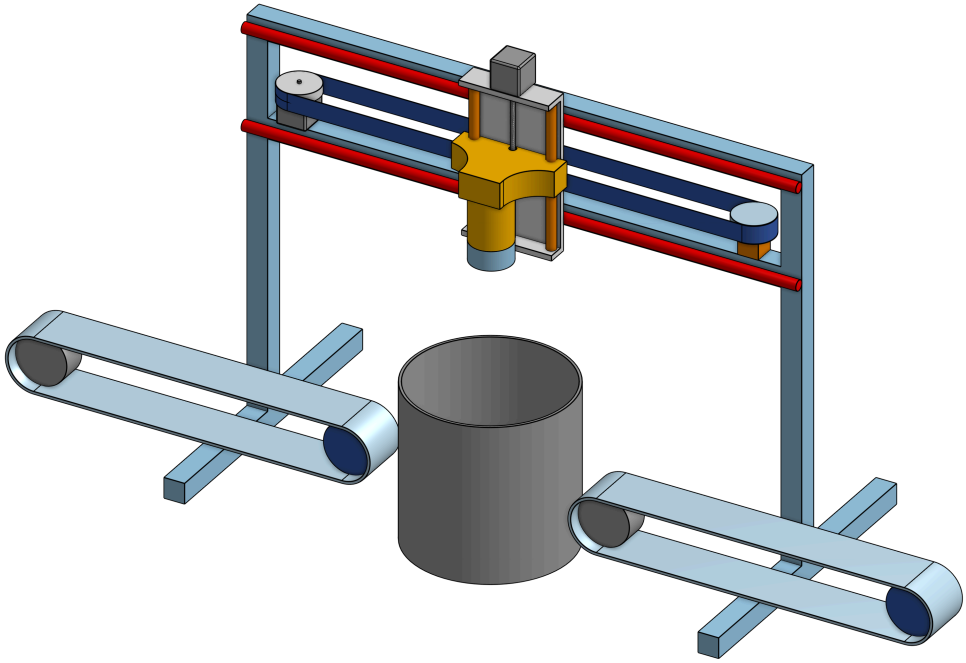


Fig. 2.1: CAD model of the Safe Dip mechanical arrangement - Perspective view

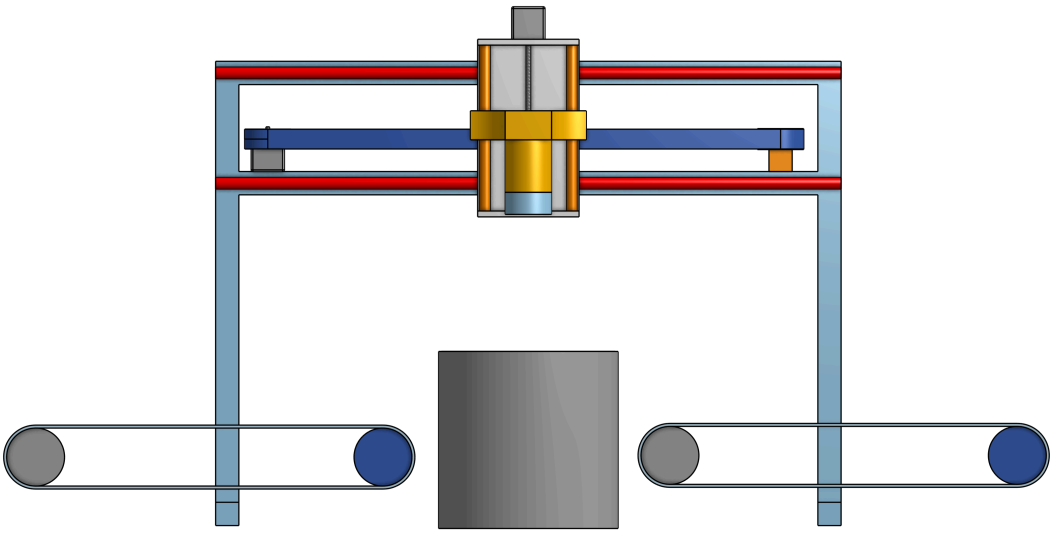


Fig. 2.2: CAD model of the Safe Dip mechanical arrangement - Front view

The CAD model in Fig. 2.1 shows the mechanical arrangement of the Safe Dip prototype. The model shows the relative placement of the horizontal axis, vertical assembly, input station, dipping station, output station, and end effector. The dimensions were checked against the selected mechanical components and the required axis travel and clearances.

The values in Table 2.1 describe the operating envelope of the prototype.

Table 2.1

Principal dimensions of the Safe Dip prototype.

Dimension	**Value**	**Design Purpose**
Overall prototype dimensions	108 cm × 60 cm × 80 cm	Defines the approximate laboratory floor space and vertical clearance required by the prototype.
Usable horizontal travel	≈ 100 cm	Allows movement between the input, dipping, and output stations.
Usable vertical travel	≈ 30 cm	Provides sufficient lifting clearance and programmed immersion depth.
Bath envelope	30 cm × 30 cm × 30 cm	Defines the approximate size of the dipping container accommodated by the prototype.
Clearance above dipping bath	≈ 5 cm	Allows the workpiece and end effector to pass above the bath during horizontal transfer.
Maximum workpiece envelope	10 cm × 10 cm	Defines the approximate maximum workpiece size supported by the prototype geometry.

2.2.2 E-Stop and Protection module

The emergency-stop and protection hardware satisfies the requirements in Table 1.4. Its purpose is to remove power from the motor drivers independently of the STM32 while treating the electromagnet as a load-holding device. This arrangement stops the hazardous motion without causing the suspended workpiece to fall into the coating bath.

A normally closed, mechanically latching mushroom emergency-stop button is wired directly into the stepper-driver power path. When the button is actuated, its contacts open immediately and remove power from the DM542TE drivers, causing the axes to coast to

a standstill and implementing a Category 0 stop. Because the stop path is hardwired and normally closed, either pressing the button or breaking a wire opens the circuit without requiring any action from the microcontroller. The button remains latched until it is manually reset, satisfying the “requires manual reset” requirement in Table 1.4.

This stop path is intentionally independent of firmware and electronic control logic. IEC 60204-1 requires a Category 0 emergency stop to operate through hardwired electromechanical components rather than depending on electronic logic [4]:.

A separate MOSFET-based electromagnet driver switches the electromagnet coil supply and is not included in the Category 0 cutoff. The electromagnet is treated as a load-holding device rather than as a hazardous actuator that must be immediately de-energized. Removing power from it during an emergency stop could create a secondary hazard by releasing the suspended ferromagnetic part into the coating bath. IEC 60204-1 requires the emergency-stop function to avoid creating such additional hazards [4]. The electromagnet therefore remains energized during the stop, and the part is released in a controlled manner only after motion has ceased.

An inline fuse holder provides overcurrent protection for the switched supply rail, while the resistor kit provides the gate and pull resistors required by the MOSFET driver and associated switching stages. A passive buzzer provides an audible indication whenever a fault or emergency-stop condition is active.



Fig. 2.3: E-stop mushroom button

2.2.3 Sensing Module

The sensing hardware provides the workpiece-presence and axis-homing feedback required for the sense-pick-dip-place sequence and satisfies the requirements in Table 1.6.

2.2.4 Power Distribution Module

This system requires three main voltage lines, one at 24 V for the stepper motor controllers, another at 12 V for the effector, and finally one of 5 V for the processing module. To that end,

two different power supplies provide the 24 V and 12 V lines, while a DC-DC buck converter generates the 5 V line.

The Mean Well LRS-350-24 is an AC-DC power supply that converts 110 or 220 Vrms into 24 V. It has four parallel terminals for the output, and it can supply a maximum current of 14.6 A.



Fig. 2.4: Mean Well LRS-350-24 AC-DC power supply

The less stringent requirements of the 12 V line means that it can be provided by an off-brand transformer, as long as it complies with the specifications. In this case, an ALITOVE DC 12V 5A Power Supply was chosen.



Fig. 2.5: ALITOVE DC 12V 5A Power Supply

Finally, for the 5 V line, as shown in Fig. 1.3, a HiLetgo buck converter based on the XL4016 chip was chosen.

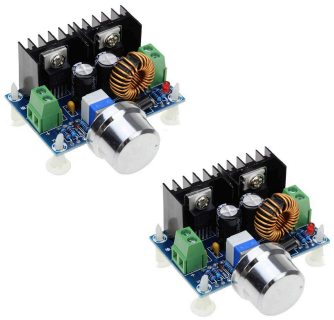


Fig. 2.6: HiLetgo buck converter

2.2.5 Processing Module

The STM32 NUCLEO-F446RE serves as the microcontroller responsible for coordinating the efforts for the sense-pick-dip-place sequence. It features an ARM Cortex-M4, 32-bit, 3.3 V microcontroller operating at a maximum frequency of 180 MHz. The STM32 family is widely adopted in embedded industrial designs due to its peripheral density, which allows for tight deterministic loops.

This board provides the necessary computing bandwidth to manage the real-time state machine while simultaneously running communication tasks. Crucially, its advanced hardware timers can be utilized in Output Compare mode to generate precise, jitter-free frequency pulses for step generation without taxing the CPU. Furthermore, its configurable interrupts can be used to immediately respond to external signals or to timer completion. The ability to delegate tasks to specialized hardware enables the Nucleo board to perform communication tasks with other peripherals.

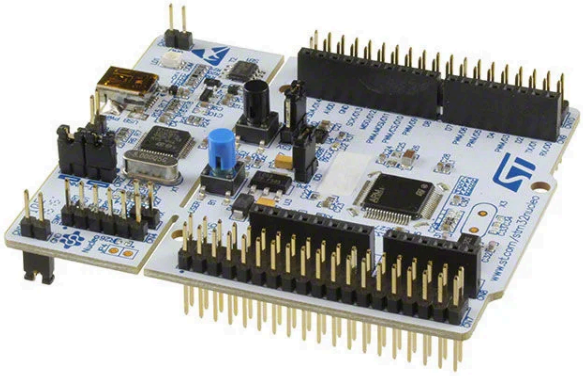


Fig. 2.7: STM32 Nucleo-F446RE microcontroller board

The STEP and DIR signals of the Nucleo board are connected directly to the optocoupler inputs of the stepper controllers, so no level-shifting buffer is used. The pins are configured as open-drain outputs that sink the optocoupler current, as described in Section 3.1.0.5.

2.2.6 Movement Module

For both the horizontal and vertical movement, a NEMA 23 stepper motor drives each axis. For this project, StepperOnline's 23HS22-2804S NEMA 23 stepper motor was chosen due to its static torque of 1.24 Nm and dynamic torque of almost 0.9 Nm at 150 RPM.

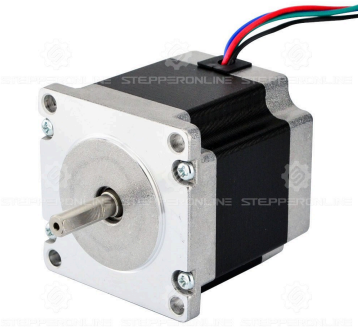


Fig. 2.8: NEMA 23 23HS22-2804S stepper motor

To drive the NEMAs using a microcontroller, a stepper motor controller is required to set their speed and direction. The DM542TE accepts a supply of up to 50 V, and in this project it is powered at 24 V and accepting 5 V signals.

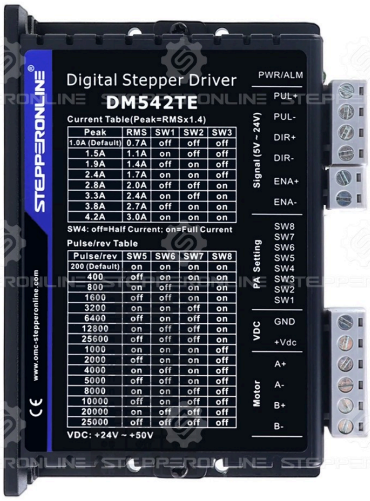


Fig. 2.9: DM542TE stepper controller

2.2.7 End Effector Module

The end-effector hardware holds the ferromagnetic workpiece throughout the pick-dip-place cycle and switches it under microcontroller control, satisfying the requirements in Table 1.8.

A Baomain BM-P 80-38 electromagnet is used as the gripping element. It holds the ferromagnetic workpiece by magnetic attraction rather than a mechanical grip, which avoids moving jaws near the coating bath and lets the part be released simply by removing coil

current. Its holding force is sized to retain the rated payload for the full duration of the dip, as required in Table 1.8, and it is driven from a single microcontroller output.

An IRLZ44N logic-level MOSFET switches the electromagnet coil from a single 3.3 V/5 V STM32 GPIO. It was chosen because its logic-level gate is compatible with the microcontroller output, and its drain-source voltage and continuous drain-current ratings comfortably exceed the 12 V electromagnet supply and coil current with transient margin, matching the MOSFET-driver requirements in Table 1.8.

A 1N4007 flyback diode is placed across the electromagnet coil to protect the MOSFET and GPIO from the inductive back-EMF generated when the coil is de-energized, addressing the "no damaging voltage transient" condition in the Table 1.8 verification.



Fig. 2.10: Baomain BM-P 80-38 electromagnet

2.2.8 Communications Module

Communications between Safe Dip and the user are divided into two categories for this project: a base case and an extension goal. For the base case, the NUCLEO-F446RE's UART peripheral, along with the ST-LINK connector on the board, will be used.

For the extension goal to communicate using Modbus-TCP, a W5500 was chosen. The W5500 is a standalone Ethernet module with a full TCP/IP stack implemented in the hardware/firmware on the chip itself. This means that the microcontroller doesn't have to run a TCP/IP stack, but instead it only has to communicate with the module via SPI.

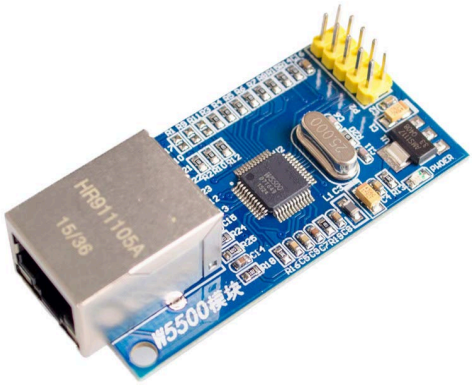


Fig. 2.11: W5500 Ethernet module

2.2.9 Electrical Schematic

Fig. 2.12 shows the complete electrical schematic of Safe Dip, drawn in KiCad. It connects the power supplies, the emergency-stop circuit, the stepper drivers, the electromagnet driver, the sensors, and the Nucleo board, and it is the reference for the wiring of the prototype.

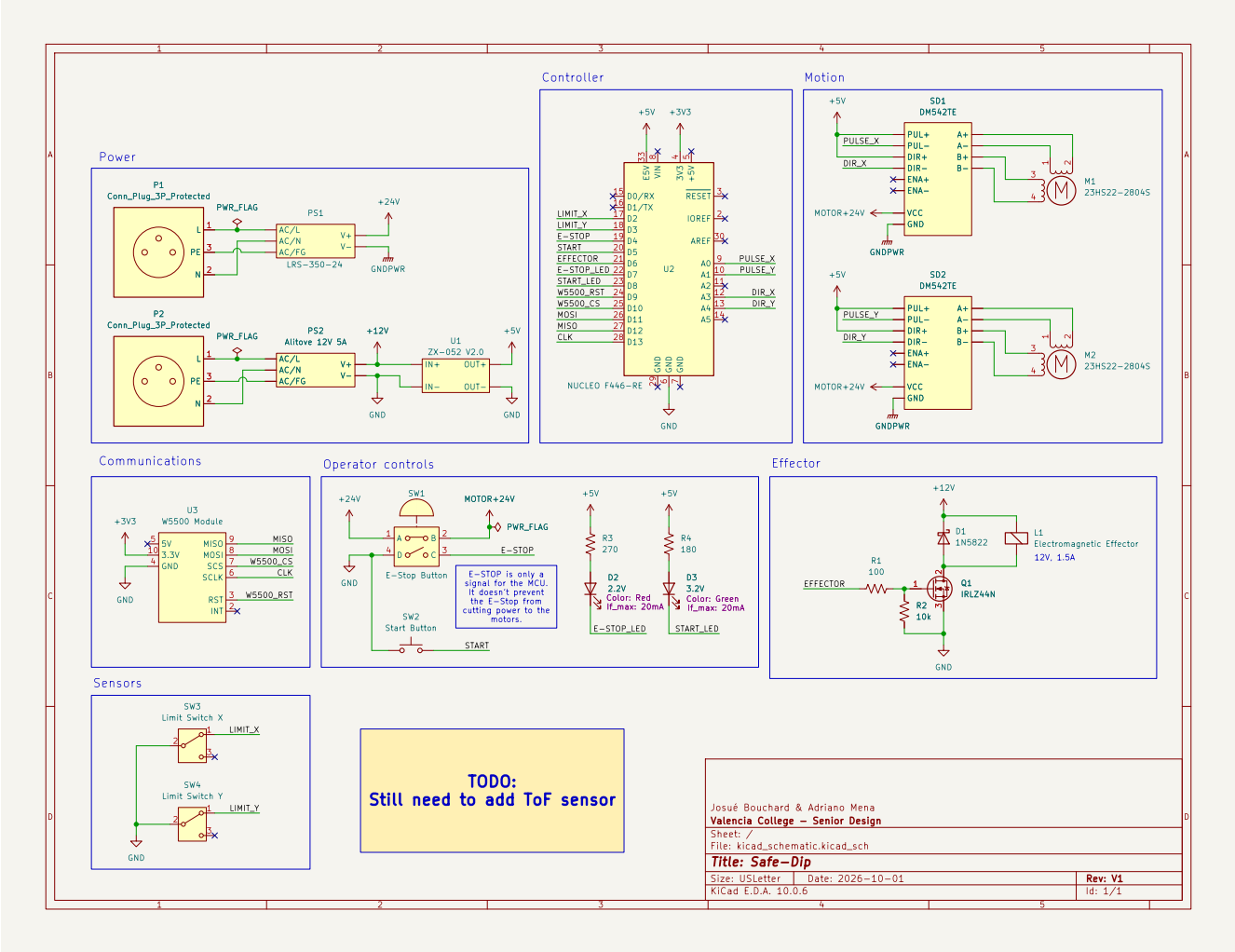


Fig. 2.12: Electrical schematic of the Safe Dip prototype.

2.3 Software and Programming Languages

The firmware and the electrical and mechanical design of the project were made with a small set of programming languages and software tools. They were chosen so that timing and memory use stay predictable on a small microcontroller, so that, wherever possible, the tools come from the component manufacturer or are free, and so that both team members can work on the same files.

The Modbus TCP library is written in C99. C is ubiquitous in embedded systems because it is simple and close to assembly code, which makes its behavior predictable and its memory use efficient. The library has no dependency on the STM32, so the same code can be built and tested on a computer, and it uses no dynamic memory. The application firmware is written in C++20 and built with the compiler set to the C++23 standard. C++ makes abstraction easier: classes and a common interface let the sequencer run any operation without knowing how it works, and features such as `constexpr` and static assertions make it easier to express constraints in the code, so that mistakes are found when the program is compiled and not while the machine is running. Strongly typed enumerations and variants also make invalid states difficult to write. The code is compiled without exceptions, run-time type information, or dynamic memory, so the language adds no unpredictable timing or memory use.

The microcontroller is configured with STM32CubeMX and the STM32Cube HAL, and the firmware is built with the STM32CubeCLT toolchain (ST Arm Clang, CMake, and Ninja), which gives the team one reproducible command-line build. clangd and clang-format keep the editor feedback and the code style consistent, the on-board ST-LINK of the Nucleo board programs and debugs the microcontroller and provides serial output, and Git and GitHub keep the history of the firmware.

Testing uses a Modbus TCP client, `madbus-cli`, to read and write coils and registers through the whole path from the computer to the STM32, an oscilloscope to check the step pulses, which come from a hardware timer and can only be confirmed by measuring the signal, and a multimeter to check the supply voltages and signal levels.

The electrical schematic is drawn in KiCad, which is free and open source and checks the wiring rules before anything is built. The mechanical parts are modeled in Onshape, which runs in a web browser so that both team members can work on the same model, and the 3D-printed parts are prepared in Bambu Studio, which previews the supports and the print time before a part is printed.

Together, these tools keep the code generated by the vendor separate from the hand-written code, rely on hardware timers and static memory for what is critical for timing, and

can be shared through a file, a repository, or an online editor, so that both team members can reproduce each other’s work.

2.4 Power Budget

The power budget of the Safe Dip prototype is given in Table 2.2, which lists the voltage, current, and power of each load, and Table 2.3 compares the resulting demand with the rating of each supply. The two stepper motors and their drivers are powered from the 24 V rail, the electromagnet from the 12 V rail, and the sensing, communications, and processing modules from the 5 V and 3.3 V rails derived from them. The figures for the motors and drivers are worst-case values, computed as the supply voltage multiplied by the rated current.

Table 2.2
Power budget for the Safe Dip prototype.

	Component	Voltage (V)	Current (A)	Unit Power (W)	Qty	Total Power (W)
Movement	Stepper motor (max usage per phase)	24	2.8	67.2	2	134.4
	Stepper motor driver	24	0.5	12	2	24
End Effector	Electromagnet	12	1.5	18	1	18
	Electromagnet driver	5	≈0.07–0.10	≈0.5	1	≈0.5
Sensing	Time-of-flight sensor	2.8–3.3	0.019	≈0.063	1	≈0.063
Communications	Ethernet module	3.3	0.132	≈0.436	1	≈0.436
Processing	STM32 (at max rate)	3.3/5	≈0.1	≈0.33	1	≈0.33
				Total operational power		**177.7 W**

Table 2.3

Maximum power available from the Safe Dip AC-DC power adapters.

Power Rail	**Power Adapter**	**Voltage (V)**	**Maximum Current (A)**	**Maximum Power (W)**
24 V	Mean Well LRS-350-24	24	14.6	350.4
12 V	12 V AC-DC power adapter	12	5	60
			Total available power	**410.4 W**

The Safe Dip prototype uses two independent AC-DC power adapters. The 24 V supply provides a maximum of 14.6 A, corresponding to approximately 350.4 W, and powers the movement subsystem. The 12 V adapter provides a maximum of 5 A, corresponding to 60 W, and powers the electromagnet and low-voltage circuitry. Together, the two adapters provide a total rated output capacity of approximately 410.4 W.

Chapter 3

Project Contribution

Summary

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3.1 Design Integration and Flow Charts

The Safe Dip control software uses a cooperative, nonblocking architecture. Rather than executing the complete pick-dip-place cycle as one blocking function, the software is divided into modules that are processed repeatedly from a single main loop: the cycle sequencer, one motion planner per axis, and the communications interface.

Each call into a module performs a bounded amount of work and returns immediately. This keeps the controller responsive to the emergency stop, the start button, and communication requests while a movement, effector operation, or dwell is still in progress.

As shown in Fig. 3.1, the software first initializes the GPIO, the W5500 controller, the Modbus TCP task, and the motion planners, and then enters a continuous main loop. At the start of every iteration, the emergency-stop and start buttons are read. If the emergency stop is pressed, the sequencer and both axes are stopped and the end effector is kept engaged (see Section 3.1.0.6). Otherwise, a press of the start button while the sequencer is not running restarts the cycle from its first operation; the sequencer is then stepped, and the motion planner of each axis is processed. The communications interface is serviced on every iteration, including while the emergency stop is pressed, so that the system state remains observable.

This structure implements cooperative multitasking. No module is allowed to block execution while waiting for an operation to finish. The motion planners continue their work over many iterations of the main loop, while the sequencer monitors the progress of the current operation.

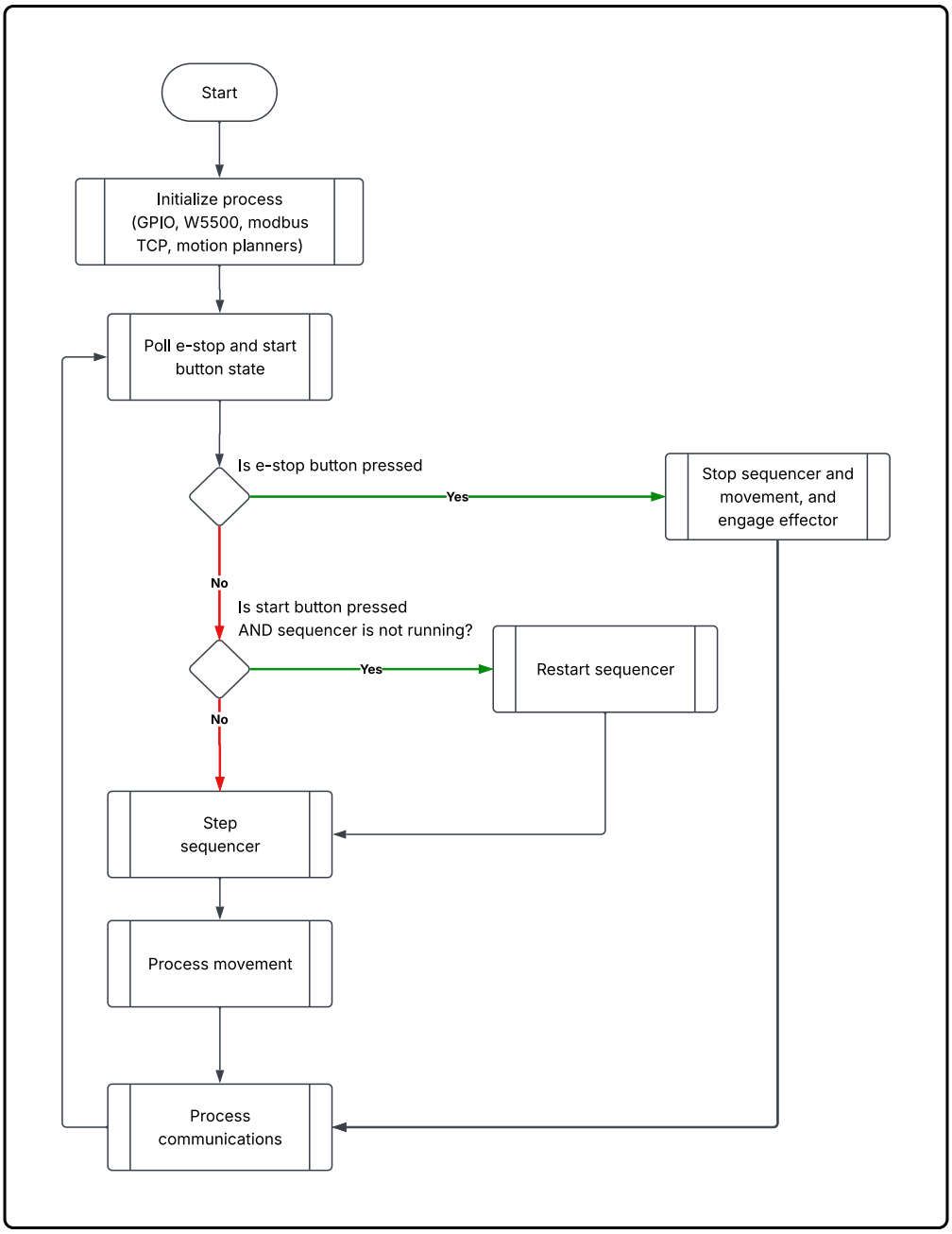


Fig. 3.1: Main software execution structure

3.1.0.1 Normal Cycle Sequence

The normal operating cycle is represented as an ordered list of nonblocking operations, shown in Fig. 3.2 and grouped into five process stages: home, pick, dip, deposit, and release. After the last operation completes, the sequence wraps around to the first, so the gantry repeats the cycle.

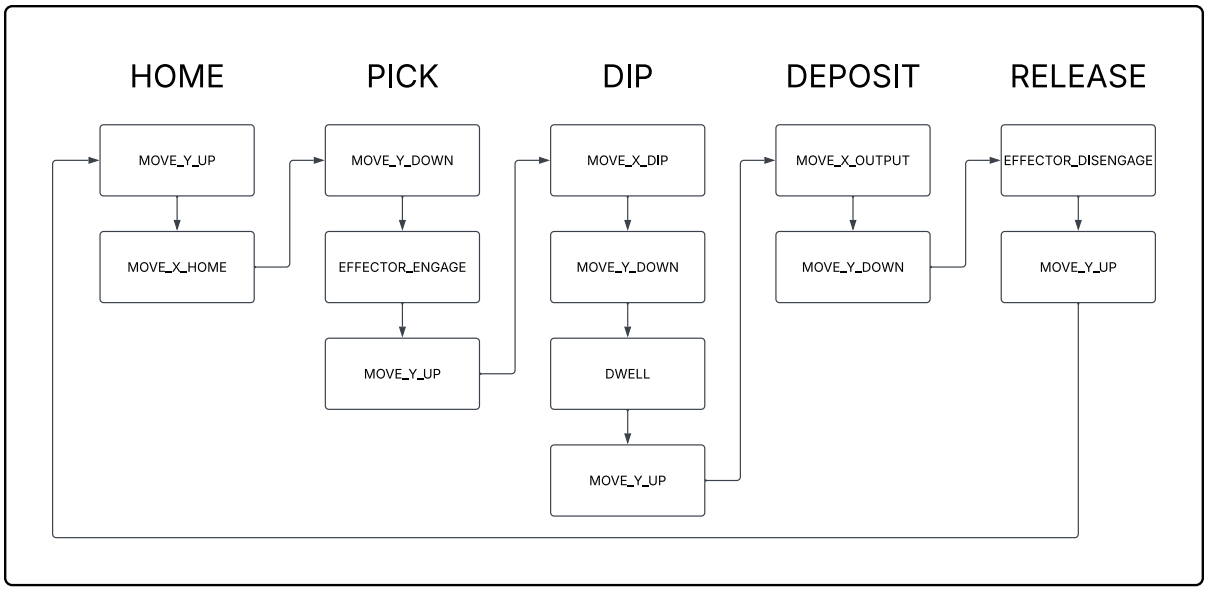


Fig. 3.2: Normal pick-dip-place operation sequence

Each operation represents an absolute objective rather than a blocking procedure. For example, `MOVE_Y_UP` requests that the vertical axis move to its configured upper position. The operation hands the target to the motion planner and returns immediately; the sequencer then keeps checking the operation until the axis reports that the target has been reached.

Two kinds of operations are planned but not yet part of the sequence. `HOME_Y` and `HOME_X` will drive each axis slowly toward its limit switch and set that point as position zero; they will run only after a start press, with the vertical axis homed first so that the end effector is raised before the gantry travels horizontally. `WAIT_FOR_PART` will hold the cycle before the pick stage until the time-of-flight sensor detects a workpiece at the input station, satisfying the requirement that no motion begins without a part present.

3.1.0.2 Operation Interface

Every entry in the sequence is an object implementing the same two-function interface:

- `start()` begins the operation, for example by giving a target position to a motion planner, and returns immediately.
- `poll()` performs any work the operation needs and reports whether the operation is still `running` or has `finished`.

The sequencer only ever calls these two functions. The correct implementation is selected at runtime through C++ virtual functions, so the sequencer needs no knowledge of which module is responsible for an operation or how that module works.

Each operation copies every configurable value it depends on when it starts. A value changed while an operation is running, for example through Modbus, therefore takes effect

the next time that operation starts and never alters a run already in progress. Table 3.1 lists the operations.

Table 3.1

Operations available to the sequencer

Operation	**`start()`**	**`poll()` reports finished when…**
Move axis	Sets the axis target position	The axis has reached the target step
Effector	Engages or releases the electromagnet	Immediately
Dwell	Records the start time and the dwell duration	The dwell duration has elapsed

This produces a clear abstraction boundary. The sequencer does not need to know which hardware timer is assigned to an axis, which GPIO pins control the stepper drivers, how many steps remain, or how the acceleration profile is calculated. Conversely, the modules do not need to know the structure of the pick-dip-place sequence, so each one can be developed and tested independently.

3.1.0.3 Cycle Sequencer

The cycle sequencer keeps an index to the current operation, a flag indicating whether that operation has been started, and a flag indicating whether the system is running. Fig. 3.3 shows one sequencer step.

If the system is not running, the step returns without doing anything. If the current operation has not been started, the sequencer starts it and returns; polling begins on the next iteration of the main loop. Otherwise, the operation is polled. While it reports `running`, the sequencer performs no further work and control returns to the main loop. When it reports `finished`, the sequencer clears the started flag and advances to the next operation, wrapping to the first after the last.

The sequencer can be stopped, which halts it in place, or restarted, which resets the index to the first operation and sets it running. The start button always uses the restart path, so the cycle never resumes from the middle of an operation whose physical state may no longer be known.

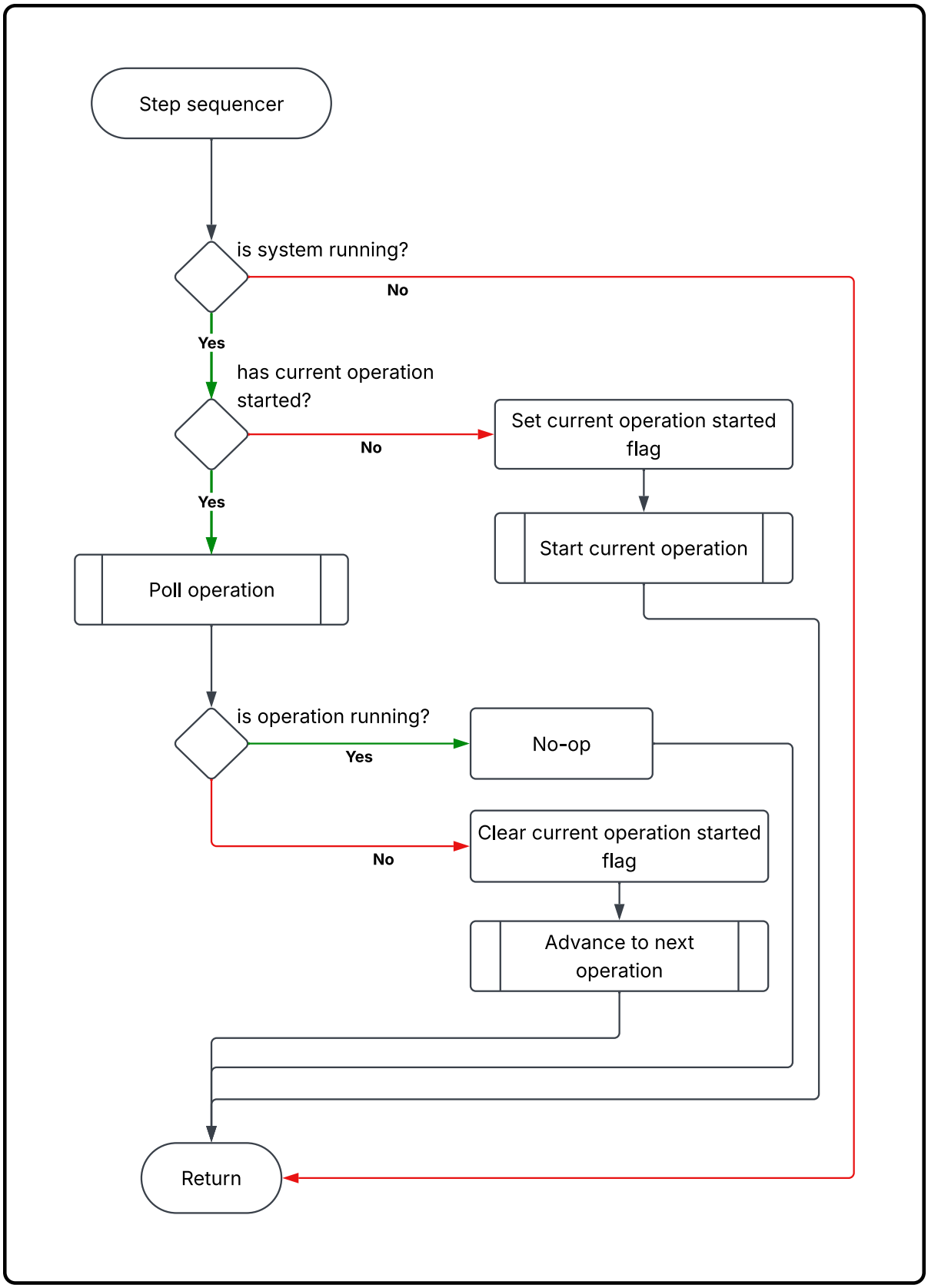


Fig. 3.3: Cooperative cycle sequencer

3.1.0.4 Motion Planner

Each axis has its own motion planner, which converts a goal into a continuously updated step rate. The planner supports two control modes. In *position mode*, used for every move in the cycle, the planner drives the axis to a target step count. In *velocity mode*, the planner holds a commanded speed; it will be used by the homing operations to jog each axis toward its limit switch.

Fig. 3.4 shows one call of the planner's `process()` function, which is made for each axis on every iteration of the main loop.

In position mode, the planner first checks whether the axis is exactly on its target step. This check runs on every call, not only once per millisecond, so that a fast-moving axis cannot step past the target unnoticed; when it succeeds, the axis is stopped immediately. Because the system tick advances once per millisecond and the main loop runs much faster, most of the remaining calls find no elapsed time and return early.

Otherwise, the planner decides whether it is time to brake. The distance needed to slow from the current speed v to the landing speed v_{land} at acceleration a is

$$d_{\text{brake}} = \frac{v^2 - v_{\text{land}}^2}{2a}$$

Once the remaining distance falls within d_{brake}, a landing flag is set and the desired speed becomes v_{land} instead of $v_{\max}$. If the axis is moving away from its target, the desired speed is zero, so the axis first brakes and then reverses. In velocity mode, the desired speed is simply the commanded speed. In both modes, the current speed is then ramped toward the desired speed by at most $a \cdot \Delta t$ and sent to the stepper driver.

The resulting motion is a trapezoidal speed profile, accelerating, cruising at $v_{\max}$, and braking, followed by a short approach at the landing speed, which lets the axis stop exactly on the target step instead of overshooting or stalling short of it. With the current tuning, $v_{\max} = 4000$ steps/s, $a = 4000$ steps/s², and $v_{\text{land}} = 100$ steps/s.

3.1.0.5 Step Pulse Generation

The motion planner only decides *how fast* each axis should turn; the individual STEP pulses are produced by hardware. TIM2 runs at 1 MHz, and each axis uses one of its channels in output-compare toggle mode. On every compare event, the timer toggles the STEP pin and an interrupt schedules the next toggle half a period later, where the half period in timer ticks is $f_{\text{timer}}/(2 \cdot |v|)$.

The interrupt also counts one step on each active edge, which gives the planner the axis position, and loads any new speed requested by the planner on each inactive edge, so

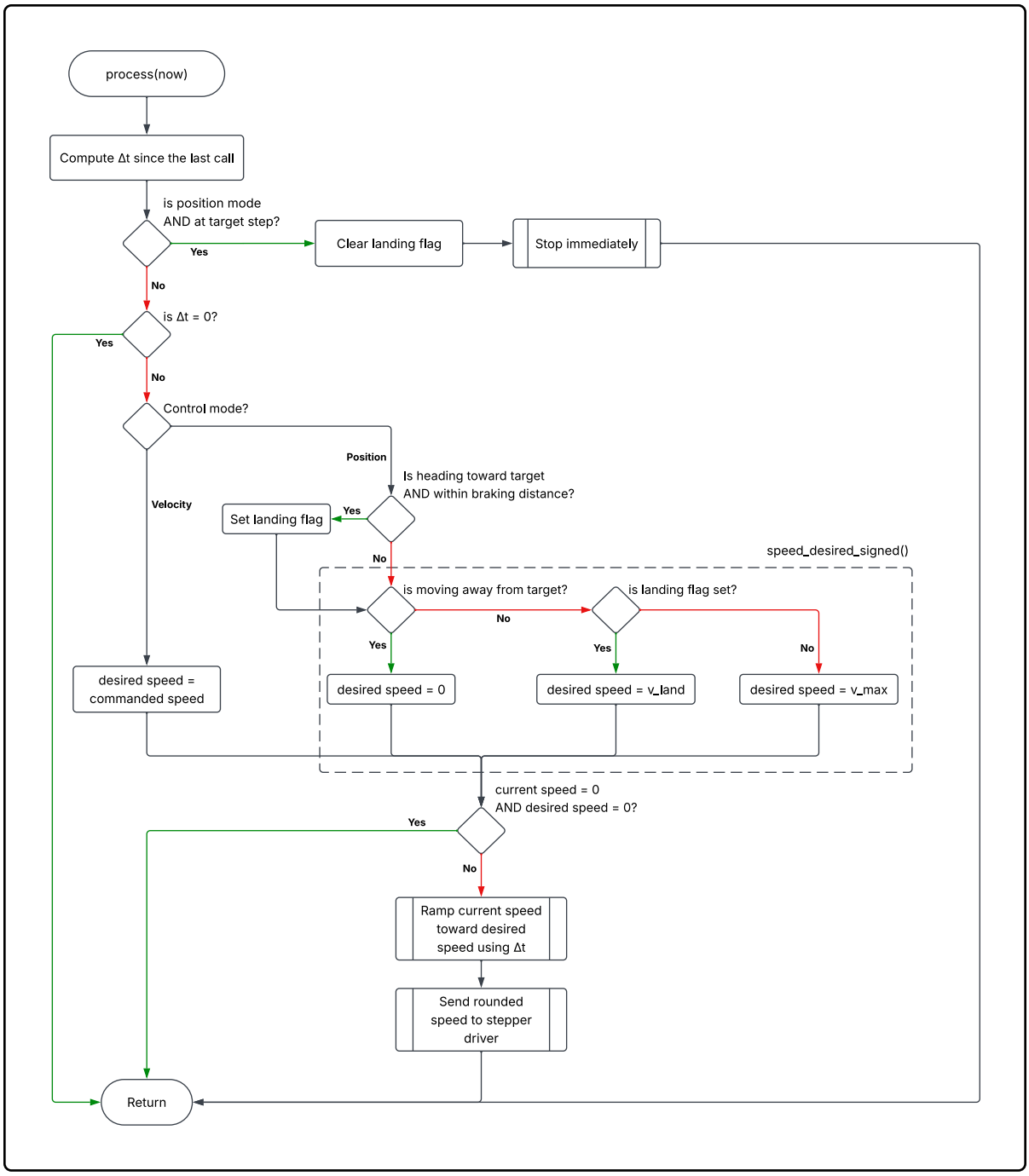


Fig. 3.4: One call of the motion planner's `process()` function. $v_{\max}$ and v_{land} are applied in the direction of the target.

that speed changes never shorten a pulse. When the requested speed is zero, the interrupt stops rescheduling itself and the channel is released.

The STEP and DIR pins are configured as open-drain outputs that sink current from the stepper drivers' optocoupler inputs, whose anodes are tied to 5 V. The microcontroller

therefore never drives these lines to 3.3 V, and the 5 V-tolerant pins interface with the drivers directly, without a level-shifting buffer.

3.1.0.6 Emergency-Stop Handling

The emergency stop is enforced first by hardware: the normally closed mushroom button opens the stepper drivers' 24 V supply directly, independently of the firmware. The firmware response supplements this path; it does not replace it.

The emergency-stop input is read at the start of every main-loop iteration. Because every module returns within a bounded time, the loop period is short enough that polling detects a press without the need for an external interrupt. While the button is pressed, the firmware stops the sequencer, stops both axes immediately by disabling their STEP timers, and keeps the electromagnet engaged so that the suspended workpiece is not dropped into the bath. The communications interface continues to be serviced.

Releasing the emergency stop does not resume the interrupted operation. A deliberate press of the start button is required, which restarts the cycle from its first operation. Because the drivers lose power during the stop, the axes may coast, and the step count no longer reflects the true gantry position. The planned homing operations will therefore run at every restart, so that each restart re-establishes a known position from the limit switches before any other motion.

Overall, this architecture combines cooperative scheduling with asynchronous peripheral operation. Hardware timers and their interrupts perform the timing-critical work, while the main loop coordinates higher-level behavior through the sequencer. This keeps the production sequence easy to understand while isolating the detailed control of the motors, end effector, dwell timing, and communications inside dedicated software modules.

3.2 Implementation

3.3 Troubleshooting

3.4 Requirement Testing and Results

3.5 Verification Summary

3.6 Final Design

Chapter 4

Non-Technical Considerations

Summary

This chapter presents the project planning and non-technical considerations associated with Safe Dip. It provides a monetary budget, a fifteen-week implementation timeline, and the distribution of responsibilities between the team members. The chapter then evaluates the environmental, health and safety, ethical, social, and sustainability aspects of the system. Particular attention is given to the hazards created by moving machinery, electrical power, and the handling of an approximately 10 kg suspended load for the fully-featured system, along with the prototype controls used to reduce those risks. Finally, the chapter discusses the project's potential to reduce worker exposure to hazardous and repetitive dipping operations while emphasizing honest reporting, maintainability, and responsible automation.

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4.1 Project Timeline

The proposed implementation schedule spans sixteen weeks. Mechanical construction and firmware development proceed in parallel so that each subsystem can be tested soon after installation. The schedule intentionally reserves the final weeks for integrated testing, debugging, refinement, and presentation preparation.

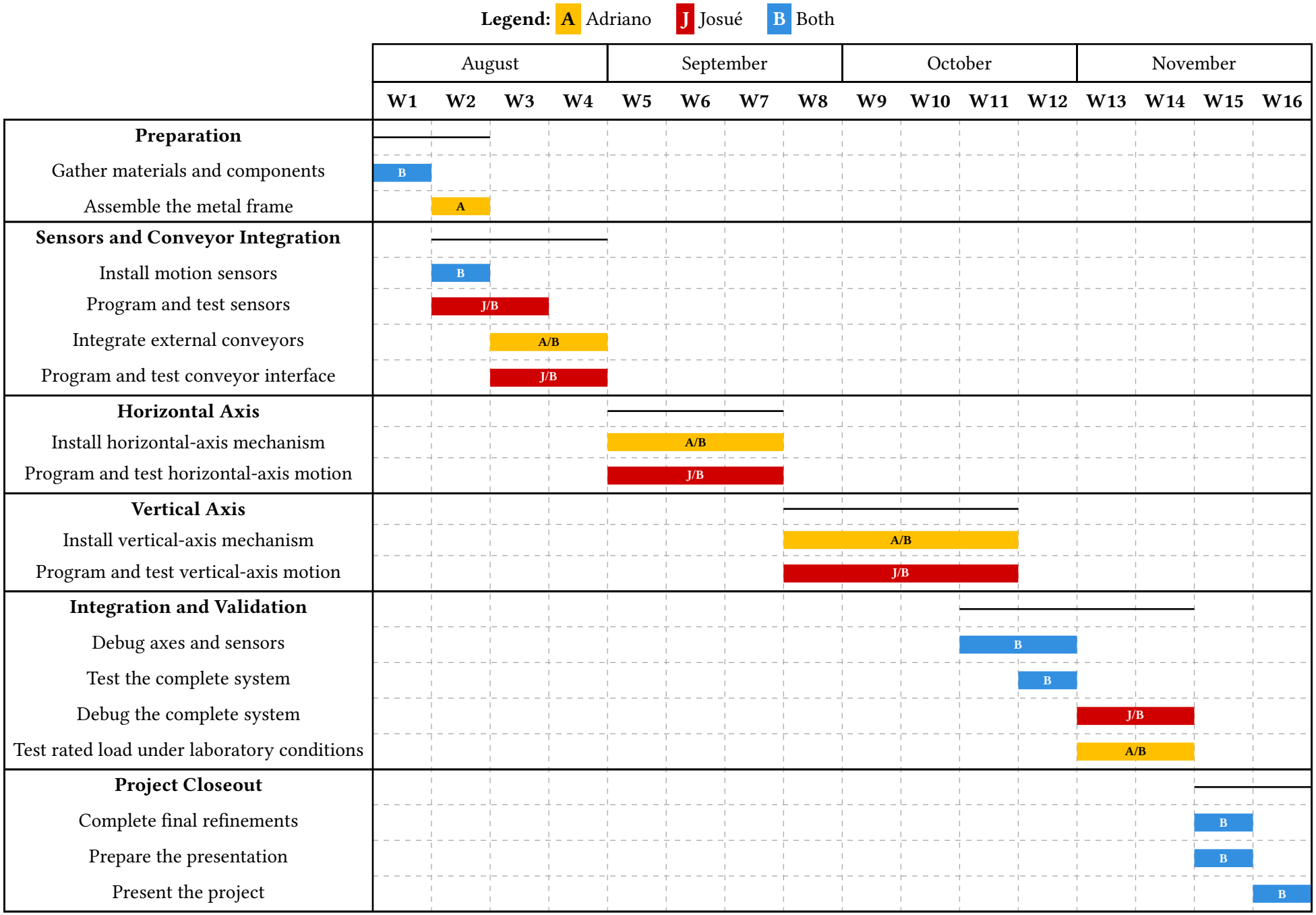


Fig. 4.1: Sixteen-week Safe Dip implementation schedule.

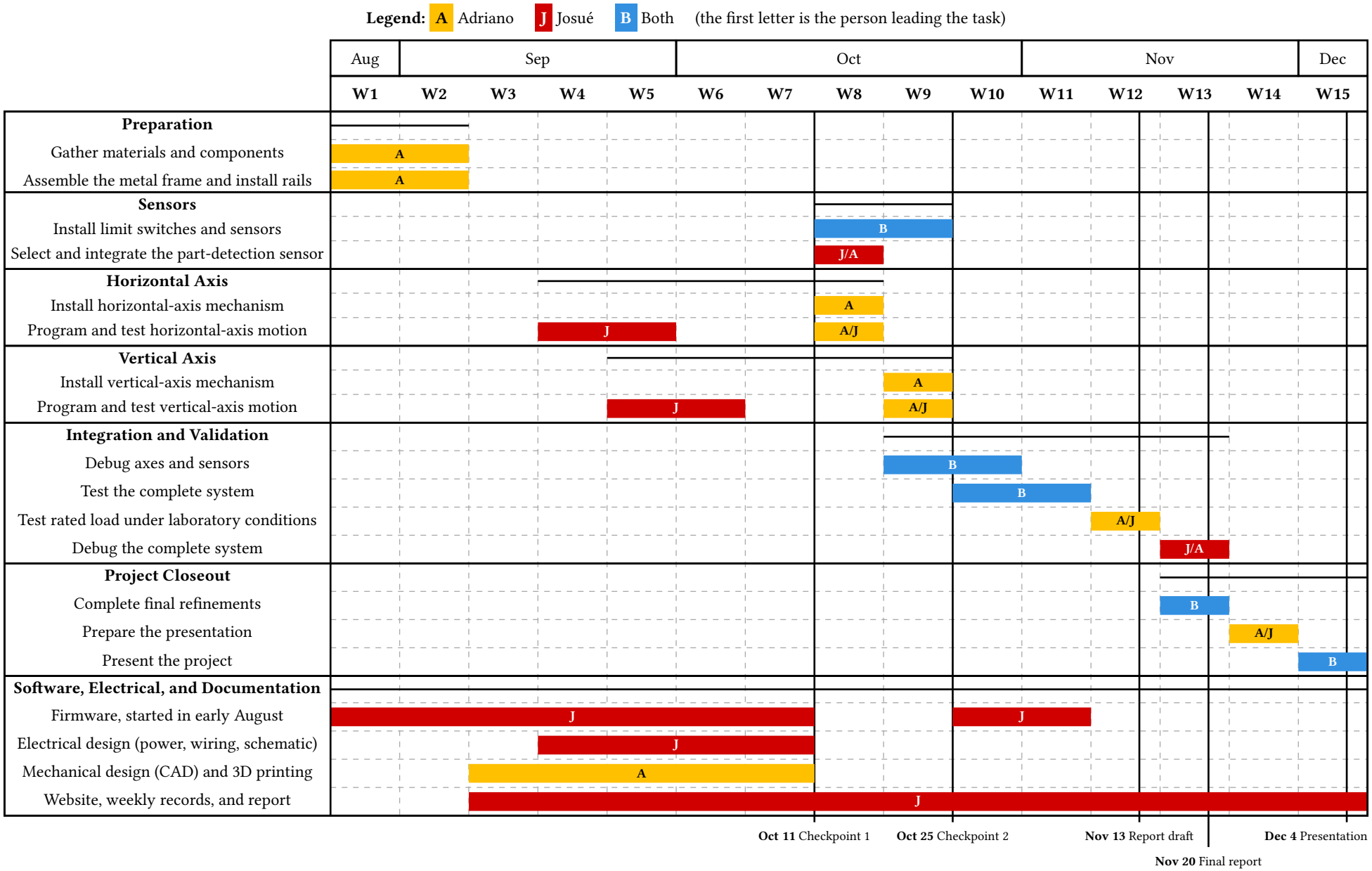


Fig. 4.2: Modified Safe Dip implementation schedule, Aug 24 – Dec 4, 2026, as of October 12, 2026. It reflects the progress to date and the current weekly task plan.

4.2 Project Expenses

The monetary budget in this section represents a preliminary estimate for the laboratory prototype. Prices were obtained from currently available commercial listings and may change as component availability, shipping costs, taxes, and final mechanical dimensions are resolved. The external conveyor is treated as supporting equipment rather than part of Safe Dip and is therefore calculated separately from the prototype total.

Table 4.1

Project component budget

Item	**Component**	**Qty.**	**Price total**
Single Output Switchable Power	Mean Well LRS-350-48	1	$35.06
Microcontroller	STM32 NUCLEO-F446RE	1	$31.00
4-channel digital buffer/line driver	SN74AHCT125N Quadruple Buffer	1	$0.80
Stepper Motor	3HS22-2804S NEMA 23	2	$26.74
Stepper Controller	DM542TE Stepper Controller	2	$42.00
Ethernet Module	W5500 Ethernet LAN Module	1	$6.50
E-Stop	1NC 1NO Red Mushroom Emergency Stop	1	$9.99
Transformer	ALITOVE Power Supply Adapter Converter Transformer	1	$14.00
Buck Converter	LM2596 DC-DC Buck Converter	1	$1.60
Electromagnet	Electromagnet Solenoid BM-P 80-38	1	$44.98
Proximity Laser Sensor	VL53L0X ToF Sensor	1	$2.98
Shaft Coupler	Shaft Coupler (6.35 x 8 mm)	2	$11.99
Limit Switch	HiLetgo KW12-3 Micro Limit Switch	1	$0.60
Rail Guide	MGN12H 300 mm Linear Rail Guide	2	$35.99
Lead Screw	TR8x2 Lead Screw (300 mm)	1	$11.99
Bearings	Pillow Block Bearing, 5/16 in (8 mm)	4	$8.49
Guide Rails	Linear Rail Slide SBR20-800 mm	2	$44.90
Belt	GT2 Timing Belt	1	$16.88

Item	Component	Qty.	Price total
Aluminum Beams	4040 T-Slot Extruded Aluminum Profile, 1000 mm	4	$79.99
Relay	One-Channel Relay Module	1	$3.70
Buzzer	Gikfun Passive Electronic Buzzer	2	$1.80
MOSFET Transistor	IRLZ44N MOSFET Transistors	4	$4.00
Terminal Block	Glarks Terminal Block Set	2	$0.26
Flyback Diode	BOJACK 1N4007 Rectifier Diode	1	$0.05
Wires	22 AWG Stranded Wire Spool	1	$12.29
Cable Wrap Insulation	Eventronic Heat-Shrink Tubing 2:1	1	$6.99
Fuse Holder	FEBRYTOLD 0-Gauge Inline Fuse Holder	1	$4.25
Resistor Kit	Upvivi Resistor Assortment Kit	1	$6.99
T-Nuts	4040 Series T-Sliding Nuts	15	$9.99
Bolt and Nut Kit	M5 Button-Head Socket Cap Screw and Nut Kit	1	$16.99
Motor Mounting	NEMA 23 Steel Mounting Bracket	4	$11.99
Nut Block	Lysee Lead-Screw Nut Block	2	$20.19
Conveyor Belt	12 V DC Electric Desktop Conveyor Belt	1	$99.51
	Total without conveyor belt (before tax)		**$525.94**
	Total with conveyor belt (before tax)		**$625.45**

4.3 Distribution of Tasks

The task assignments shown in Fig. 4.1 identify a primary owner for each activity, but they do not prevent either team member from assisting with work outside that assignment. Mechanical and software tasks are deliberately paired during integration so that installation, programming, and validation decisions are made jointly. Table 4.2 gives the preliminary distribution, and Table 4.3 gives the current distribution, after the changes described below.

Table 4.2

Preliminary distribution of project responsibilities.

Team member	**Primary responsibilities**
Josué Bouchard	Embedded firmware; time-of-flight and limit-switch interfacing; horizontal- and vertical-axis motion control; supervisory state machine; Modbus TCP integration; system debugging; and software-oriented verification.
Adriano Duque Mena	Mechanical frame assembly; installation of rails, transmissions, motors, sensors, and external conveyor interfaces; mechanical troubleshooting; fabrication support; and preparation of rated-load testing.
Both	Component procurement; electrical and mechanical integration; emergency-stop validation; complete-system testing; final refinement; report preparation; and presentation.

As the build has progressed, responsibilities have shifted in several areas. Table 4.3 shows the distribution as it stands now.

Table 4.3
Current distribution of project responsibilities.

Team member	**Primary responsibilities**
Josué Bouchard	Embedded firmware; selection and interfacing of the workpiece-detection sensor and interfacing of the limit switches; horizontal- and vertical-axis motion control; supervisory state machine; Modbus TCP integration; electrical design, including the power distribution and wiring; system debugging; software-oriented verification; project documentation; and preparation of most of the report.
Adriano Duque Mena	Mechanical design in CAD, including the vertical assembly; mechanical frame assembly; installation of rails, transmissions, motors, and sensors; fabrication of custom parts; procurement of the mechanical components; mechanical troubleshooting; preparation of rated-load testing; and preparation of the mechanical parts of the report.
Both	Procurement of the remaining components; electrical and mechanical integration; emergency-stop validation; complete-system testing; final refinement; and presentation.

The main changes are:

- The external conveyor interfaces were removed from the project scope, so the conveyor activities in Fig. 4.1 no longer apply.
- The choice of workpiece-detection sensor is still being decided, with the time-of-flight sensor as one of the options.
- Electrical design and project documentation became part of Josué's responsibilities.
- Mechanical design in CAD and the procurement of the mechanical components became part of Adriano's responsibilities.
- Report preparation, previously shared, is now split: Josué writes most of the report, and Adriano contributes the mechanical parts.

Two factors drove these changes. The first is the dependencies between tasks: the electrical design depends on the firmware, which defines the controller's pin assignments and the driver interfaces, and procurement depends on the mechanical design, which determines which parts are needed. The second is availability and timing: when a task became ready, it

was taken by the team member who could start it first, rather than waiting for the originally assigned owner.

4.4 Health and Safety

Safety is the central motivation for Safe Dip. The system is intended to remove the operator from direct interaction with a repetitive dipping task and, in the full application, from proximity to hot and viscous coating material. The prototype will be evaluated under laboratory conditions without process heat, handling a workpiece of approximately 1 lb. However, the system was designed in preparation for handling 10 kg workpieces, which will present meaningful mechanical, electrical, and suspended-load hazards.

The movement emergency stop is implemented as a Category 0 stop of the movement subsystem. Actuating the normally-closed mushroom button removes power from both stepper-motor drivers immediately. The button requires a twist action to reset, and motion must remain inhibited until the button has been reset and the operator deliberately restarts the system. The electromagnet is powered from a separate supply that is not interrupted by the motion emergency stop so that an attached workpiece is not intentionally released when axis power is removed.

Table 4.4

Primary prototype hazards, controls, and residual limitations.

Hazard	**Prototype control**	**Residual limitation**
Unexpected axis motion	Normally-closed, twist-reset mushroom emergency stop removes power from both motor drivers. Firmware prevents automatic restart after a stop event.	The emergency stop does not replace safe setup procedures or physical separation from the moving gantry.
Pinch and crush points	Motion is limited to controlled laboratory tests, and personnel must remain clear of the rails, leadscrew, belt transmission, and carriage while the system is enabled.	The current prototype does not claim the protection provided by a fully guarded industrial cell.

Hazard	Prototype control	Residual limitation
Suspended workpiece	The magnet remains powered during a motion emergency stop. Retention is tested before complete-cycle operation, and no person may stand beneath the workpiece.	A conventional electromagnet does not retain the workpiece after a total loss of magnet power. This is a prototype limitation and must not be described as fail-safe load retention.
Electrical fault or short circuit	Fuses are installed in the prototype power branches, and exposed connections are insulated and secured before operation.	Fuse ratings and conductor sizes must be finalized from the actual branch currents and selected hardware.
Loss of position reference	Normally-closed limit switches provide homing and boundary references, while the time-of-flight sensor verifies workpiece presence before the cycle begins.	Sensor detection does not make entry into the machine workspace safe while motion is enabled.
Thermal or coating exposure	No heated coating material is used during prototype acceptance testing.	The prototype does not validate operation at the intended industrial temperature or with a production coating medium.

The conventional electromagnet creates an important distinction between an emergency stop and a complete power failure. The separate supply preserves holding force when only the motion subsystem is stopped, but a failure of the magnet supply may still release the load. This limitation must remain visible in the operating instructions, test procedure, and final evaluation of the prototype.

4.5 Ethical Aspects

The ethical responsibility of the project team extends beyond demonstrating that the mechanism can complete a pick-dip-place cycle. Because Safe Dip is presented as a safety-oriented system, performance and safety claims must be based on documented tests rather than design intent alone. In particular, the report must clearly distinguish between the intended 10 kg,

elevated-temperature product concept and the laboratory prototype, which is tested without heat and with several safeguards that remain incomplete relative to an industrial installation.

The team will report failed tests, unresolved hazards, and prototype limitations honestly; will not describe the conventional electromagnet as retaining the load after complete power loss; and will not claim compliance with an industrial safety standard unless the relevant architecture and validation have actually been completed. Libraries, technical references, component data, figures, and external designs used during development will be credited appropriately. Safety decisions will take priority over cycle speed, appearance, or demonstration convenience.

Safe Dip does not require the collection of personal information. Operational data communicated through the serial or Modbus TCP interface is limited to machine state, sensor values, commands, and fault information. Any future logging system should collect only the information needed for diagnostics and process traceability.

4.6 Environmental Aspects

The laboratory prototype is not intended to use a heated bath or process chemicals. Its direct environmental impact is therefore primarily associated with electricity consumption, packaging waste, metal and plastic fabrication waste, and the eventual disposal of electronic components. Scrap aluminum, steel, wire, and electronic assemblies should be separated whenever practical so they can be reused or sent through an appropriate recycling stream rather than discarded as mixed waste.

The intended industrial application may use wax, resin, or another coating medium at elevated temperature. Because that operating condition is outside the current prototype test envelope, this report does not claim validation of spill control, fume generation, coating disposal, or thermal energy consumption. Those subjects must be evaluated using the selected coating material and production process before the system is deployed outside laboratory conditions.

4.7 Social Aspects

Safe Dip is intended to reduce worker exposure to a repetitive and potentially hazardous manual process. Removing the operator from the immediate dipping area may reduce physical strain and, in the full-temperature application, reduce the risk of burns and contact with hot coating material. A lower-cost gantry system may also make basic automation more accessible to small and medium manufacturers that cannot justify a full industrial robotic cell.

Automation can also create concern about the replacement of manual labor. The system is better understood as transferring the operator away from direct handling and toward setup, supervision, inspection, maintenance, and process control. Its value should therefore be evaluated not only through throughput, but also through changes in worker exposure, training needs, system reliability, and the quality of the resulting work environment.

4.8 Sustainability

The design supports long-term maintenance by using a modular frame and commercially available motors, drivers, sensors, power supplies, and communication hardware. A failed subsystem can therefore be repaired or replaced without discarding the complete machine. The aluminum structure, rails, and mechanical transmission can also remain in service when the controller, sensors, or end effector are upgraded.

The conventional electromagnet is a practical prototype choice, but it consumes power continuously while holding a workpiece. The control system should therefore energize it only when retention is required and should avoid unnecessary idle holding periods. This energy-use tradeoff should be considered when comparing the prototype with a future electro-permanent or mechanical holding solution.

Repeatable control of immersion depth and dwell time may reduce coating inconsistency and material waste in a future process implementation. That benefit is not demonstrated by the present non-heated prototype and should remain a design objective until it can be measured using the selected coating medium.

Chapter 5

Conclusion

Summary

This chapter concludes the Safe Dip report by reviewing the purpose and architecture of the system, identifying the expected contribution of the laboratory prototype, and outlining improvements that may be considered in future development.

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5.1 Summary and Conclusion

Safe Dip is a safety-oriented pick-dip-place system designed to reduce worker exposure to the hazards associated with manual hot-melt coating operations. The design uses a two-axis Cartesian gantry to detect, acquire, transport, dip, and release a workpiece without requiring the operator to remain near the dipping area.

The system is controlled by an STM32 microcontroller that coordinates the stepper-driven horizontal and vertical axes, the time-of-flight sensor, limit switches, electromagnet, emergency-stop input, and communication interfaces. Serial communication provides the baseline operator interface, while Modbus TCP through the W5500 module allows system states, sensor information, and fault conditions to be observed by external equipment.

Safety is a central consideration in the design. A normally-closed, twist-reset emergency-stop button immediately removes power from the movement subsystem, while the conventional electromagnet remains powered from a separate supply so that actuating the movement emergency stop does not intentionally release the suspended workpiece. Fuses are also included in the prototype power branches to reduce the consequences of electrical faults.

The laboratory prototype will be tested without a heated coating bath and will handle a workpiece of approximately 1 lb under controlled conditions. Successful testing would demonstrate the feasibility of the control, sensing, communication, movement, and emergency-stop architecture, but it would not validate operation at the intended industrial temperature. The conventional electromagnet also cannot provide load retention during a complete loss of magnet power, which remains an important prototype limitation.

Overall, Safe Dip proposes a lower-cost and modular alternative to manual dipping and more expensive industrial robotic systems. The project provides a foundation for determining whether the design can later be developed into a guarded, temperature-rated, and industrially deployable machine.

5.2 Suggestions for Future Work

Future development should first address the safeguards required for operation near a heated coating bath. These additions may include a transparent physical enclosure, interlocked access points, independent overtemperature protection, spill containment, and visual and audible status indicators.

The main focus of future improvents would be to harden the system by providing thermal protection to the mechanism. The conventional electromagnet may also be replaced by an electro-permanent magnet, mechanically locking gripper, or another end effector

capable of retaining the workpiece during a complete power failure. Additional sensing could verify that the workpiece has been successfully acquired before movement begins.

Further improvements may include closed-loop position feedback, enhanced fault monitoring, a dedicated operator interface, and standardized communication with upstream and downstream conveyor equipment. After these safeguards are implemented, the system could be evaluated using the intended coating material and operating temperature to measure retention, positioning accuracy, dip consistency, and long-term reliability.

Also, because the electromagnet is deliberately held during an emergency stop, it depends on its 12 V supply remaining available. A planned future improvement is to add a small backup battery on the electromagnet rail, sized to hold the coil energized for a few minutes in the event of a total power loss, so that a suspended part is not dropped when mains power is lost rather than merely when the E-stop is actuated

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Appendix A - Programming Code

Appendix B - Applicable Engineering Codes and Standards

The codes and standards in Table B.1 were identified as relevant references for the design and development of Safe Dip. Their inclusion does not represent a claim of certification or complete compliance. Formal compliance would require evaluation of the finished system against the complete requirements of each applicable document.

Table B.1
Engineering codes and standards applicable to the Safe Dip project.

Code or Standard	**Organization**	**Explanation**	**Relevance to Safe Dip**
IEEE Code of Ethics	Institute of Electrical and Electronics Engineers	Defines ethical duties related to safety, honesty, competence, and public welfare.	Supports safe design, truthful reporting, and clear disclosure of prototype limitations.
NSPE Code of Ethics for Engineers	National Society of Professional Engineers	Requires engineers to prioritize public safety and communicate technical information honestly.	Applies to safety decisions, testing, risk disclosure, and performance claims.
ISO 13850:2015	International Organization for Standardization	Defines requirements and design principles for machinery emergency-stop functions.	Guides the latching E-stop, immediate motion cutoff, manual reset, and restart inhibition.
ISO 14118:2017	International Organization for Standardization	Addresses prevention of unexpected machine start-up.	Requires a deliberate restart action after the E-stop is reset.

Code or Standard	Organization	Explanation	Relevance to Safe Dip
NFPA 79	National Fire Protection Association	Covers electrical equipment, wiring, protection, and controls for industrial machinery.	Guides internal wiring, branch protection, grounding, control circuits, and E-stop wiring.
NFPA 70 National Electrical Code	National Fire Protection Association	Defines requirements for safe electrical installations and supply connections.	Guides power connections, wiring, fuses, grounding, and electrical fault protection.

Appendix C - Datasheets

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Appendix D - Calculations

Leadscrew calculations

The vertical-axis capacity is determined by combining the leadscrew kinematics with the motor's torque–speed characteristic. For a leadscrew with lead L, the required rotational speed is obtained from the desired linear speed using Fig. D.1. The motor's torque–speed curve then provides the available torque $T(\omega)$ at that operating speed. Finally, Fig. D.2 converts the available torque into the maximum safe payload.

$$v = nL$$

Fig. D.1: Relationship between leadscrew rotational speed and linear speed.

$$F_{\text{axial}} = \frac{2\pi T(\omega)\eta}{L}$$
$$m_{\max} = \frac{F_{\text{axial}}}{g} = \frac{2\pi T(\omega)\eta}{Lg}$$
$$m_{\text{safe}} = \frac{m_{\max}}{S} = \frac{2\pi T(\omega)\eta}{SLg}$$

Fig. D.2: Maximum safe payload mass produced by the vertical leadscrew.

In these equations, L is the leadscrew lead, defined as the axial distance traveled by the nut during one complete revolution [5]. The term η represents the mechanical efficiency of the leadscrew and accounts for frictional losses, while S is the selected safety factor. For this design, a safety factor of $S = 2$ is applied to account for uncertainties in loading, acceleration, drivetrain losses not captured by the nominal efficiency, and variations between the theoretical and available motor torque.

The torque used in Fig. D.2 should be obtained from the motor's torque–speed curve at the intended operating speed. The static holding torque should not be used for a moving-axis calculation because the available motor torque decreases as rotational speed increases.

Appendix E - Team Biographies



Josué Bouchard is originally from Argentina and is completing his Bachelor's degree in Electrical and Computer Engineering Technology at Valencia College. His interest in engineering began with an Arduino board and grew from a fascination with using software to interact with the physical world. After beginning his studies in Computer Engineering at the Universidad de Buenos Aires, he transferred to Valencia College, where he combined his theoretical background with hands-on experience in electronics, microcontrollers, and embedded systems. Josué also enjoys helping others understand technical concepts through his work as a laboratory aide. After graduation, he hopes to pursue a career in embedded systems and eventually work in aerospace or robotics.



Adriano Mena Duque de Estrada was born in Colón, Matanzas, Cuba, and immigrated to the United States with his family in 2016, crossing the border with very little and no knowledge of English. With faith and perseverance, he is now completing his Bachelor's degree in Electrical and Computer Engineering Technology at Valencia College, having switched from Aerospace Engineering after realizing it wasn't the right fit — though designing spacecraft remains a personal dream. Outside of coursework, Adriano enjoys strategy video games, spends time with his dog and bearded dragon, and loves building electrical projects, which is what ultimately drew him into this career path.